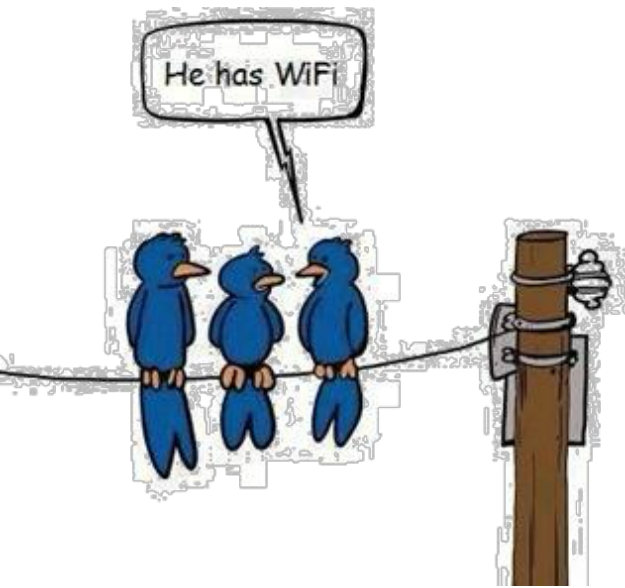


ECSE 4560/ 6560

Modern Communication Systems

Lecture 14: OFDM Revision and MIMO Intro

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **OFDM Revision**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
- HW/Lab 4 due Wed **Oct 22** (Late due Friday Oct 24).
- Exam 1: **Friday Oct 24** -- practice problems coming soon
- Exam 2: Friday Nov 21
- Final Project Presentation and Demos: Dec 9 and Dec 12
- Project Report and peer evaluation: Finals exam week

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - MIMO Chapter 6
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class

- OFDM Revision
 - HW/Lab 3, 4
 - SFO and Residual Phase correction (ECSE6560 topic)
- MIMO Intro
 - MIMO in WiFi Standards
 - MIMO Intro
 - MIMO vector representation

OFDM: Putting it Together

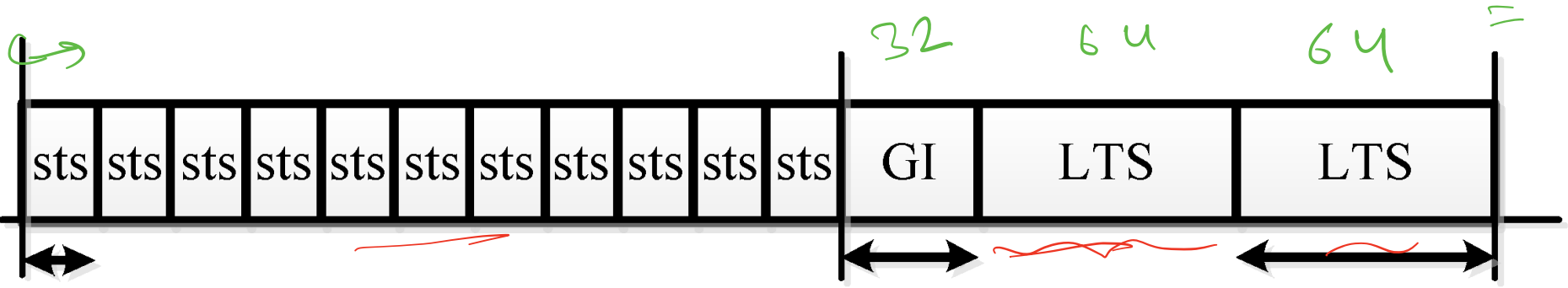
At TX:

- Create preamble symbol from training sequence (Uses BPSK)
- Repeat preamble symbol:
 - ~10 times STS for coarse packet detection
 - 2 times LTS for fine packet detection
 - 2 times LTS for CFO estimation
 - 2 times LTS for channel estimation
 - Add CP for LTS preamble
- Create data symbol from:
 - Data bits (Uses BPSK, QPSK, M-QAM)
 - Pilot bits (Uses BPSK)
- Add cyclic prefix to data symbols.

WiFi Example: STS and LTS

1 ts - t
1 ts - f ↓

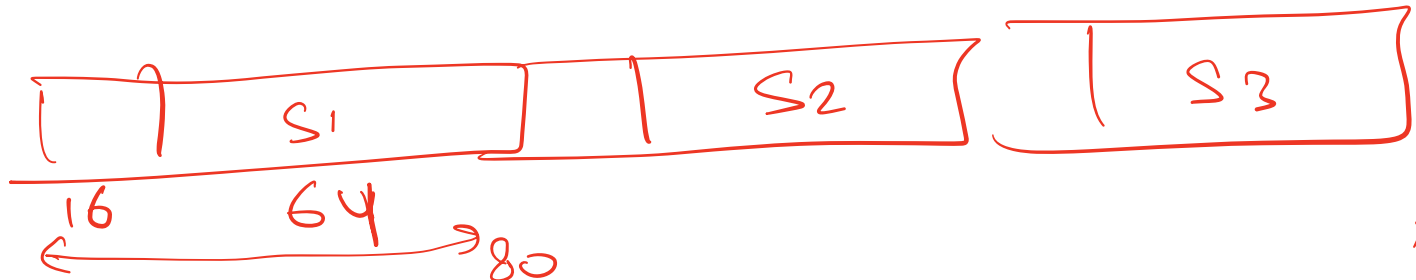
16 x 10



0.8 μs
1.6 μs
3.2 μs
160
2.5 LTS

⇒ 160

160 + 160 = 320 preamble



... later
x 50 = 4000

OFDM: Putting it Together

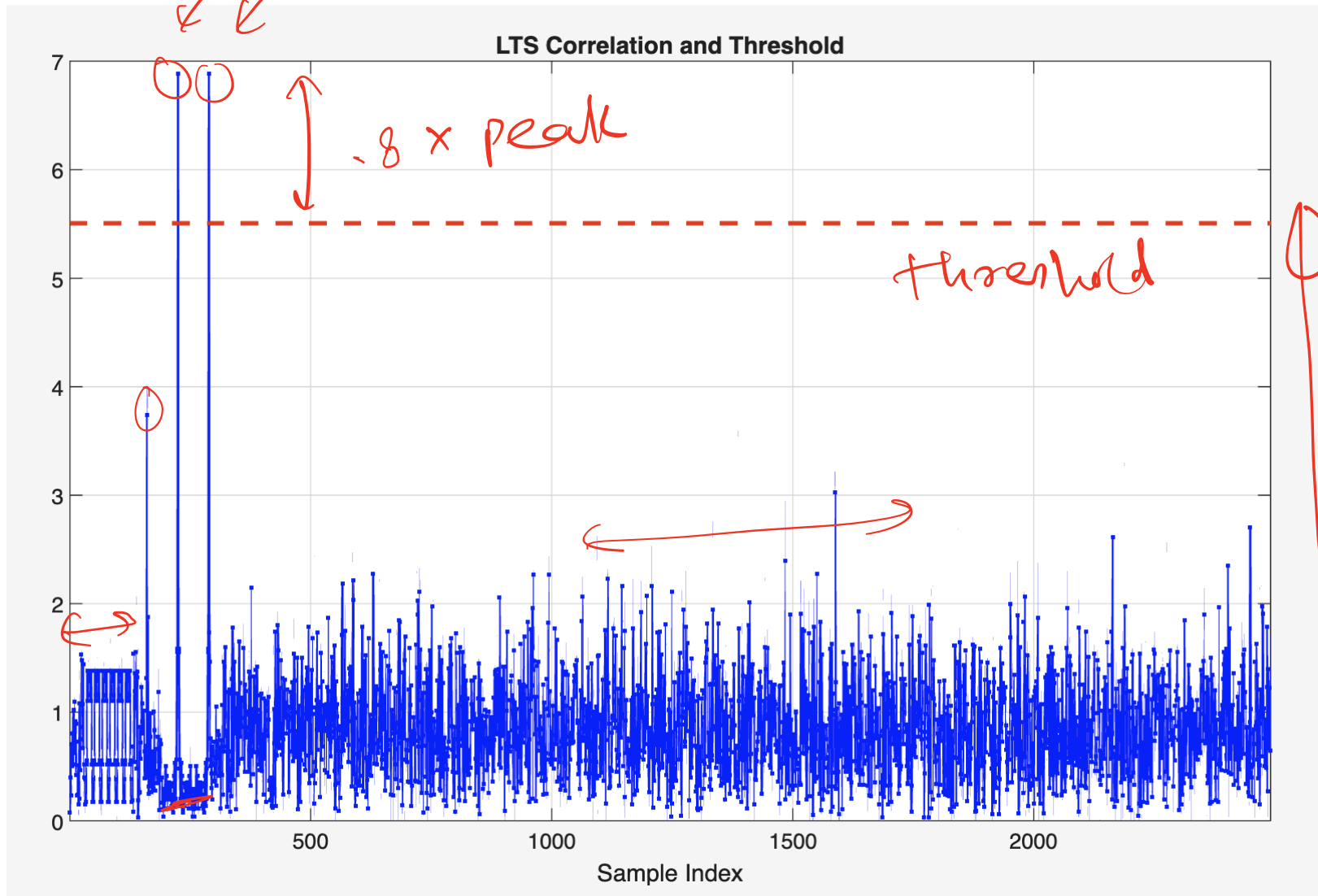
At RX (HW 4):

- Detect beginning of packet. ← mostly LFS, correlation
- Estimate & correct for CFO. ← cross
- Jump $\approx CP - 4$ samples into symbol to avoid ISI
- Estimate the channel. ← Auto-correlation with Schmidl-Cox $\Rightarrow$ Better!
- For each subsequent data symbol:
 - Remove CP
 - Take FFT of Size N
 - Correct for channel by dividing with $\tilde{H}(f)$
 - Use linear regression to estimate residual CFO and SFO with pilots
 - Correct for residual CFO and SFO
 - Decode Bits

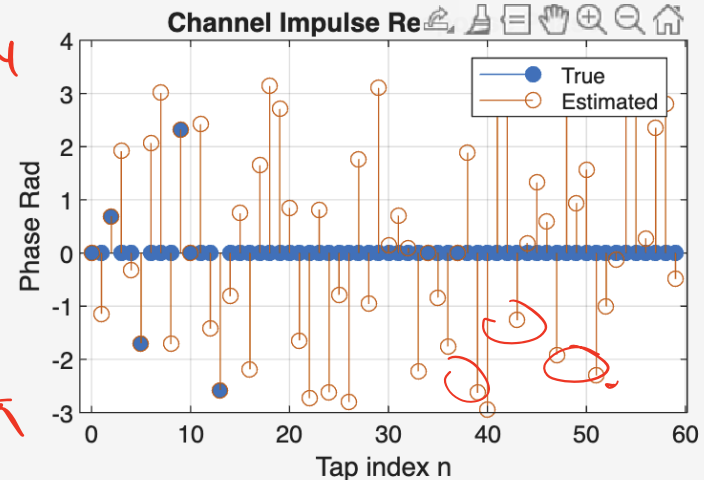
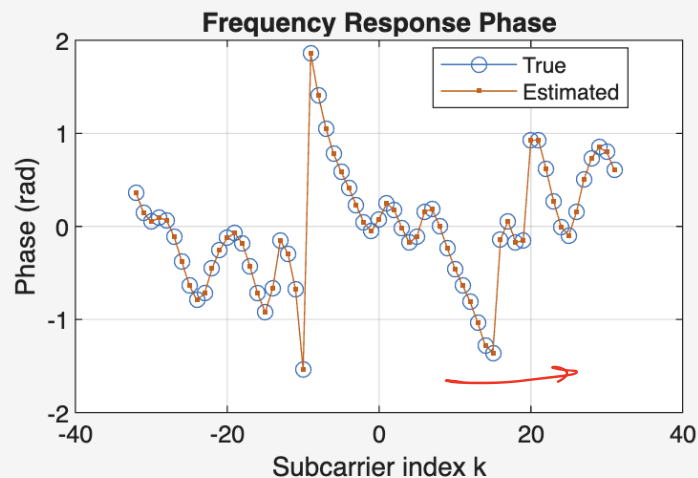
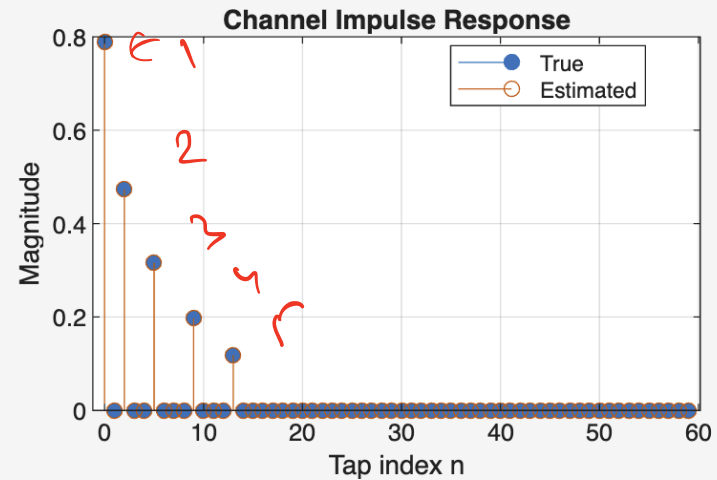
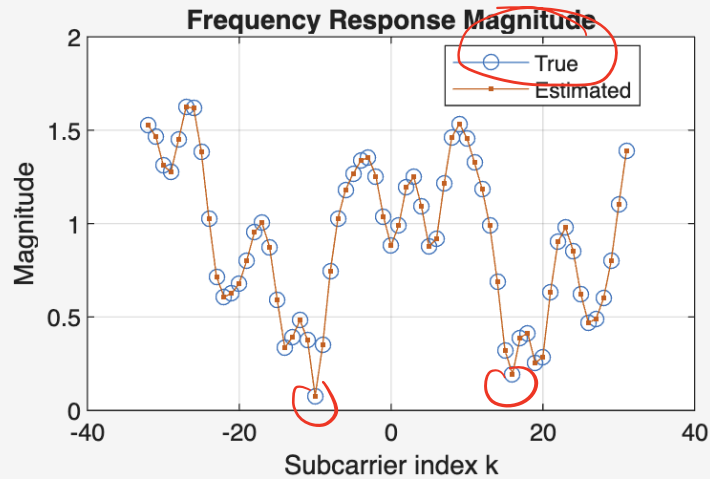
BER

Packet Detection

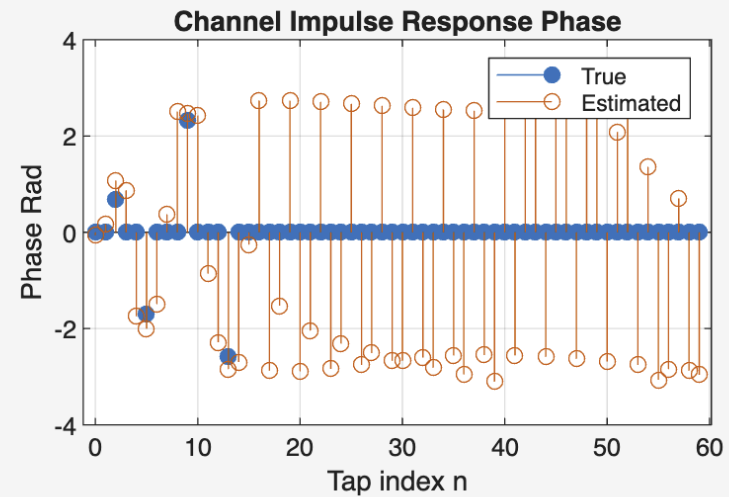
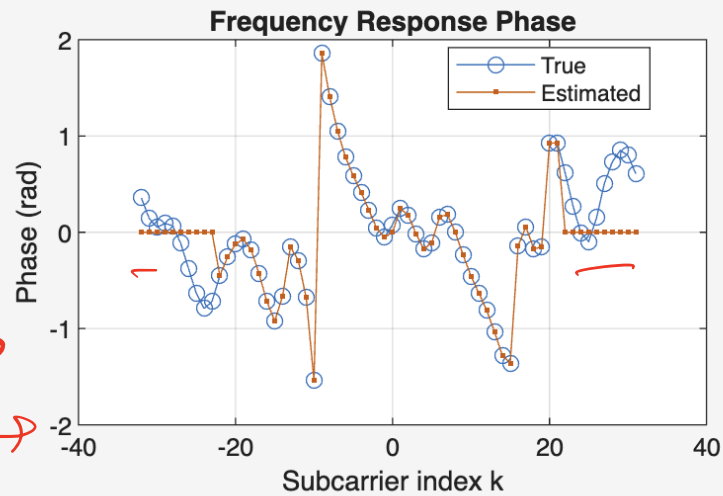
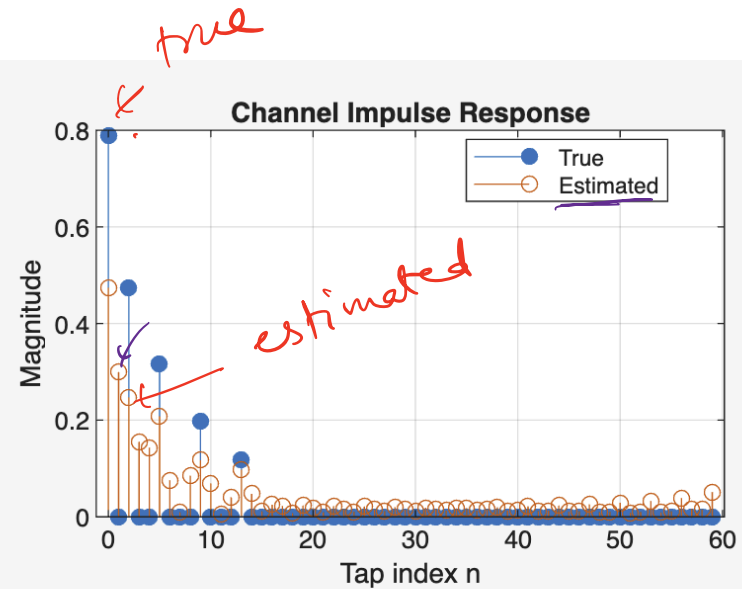
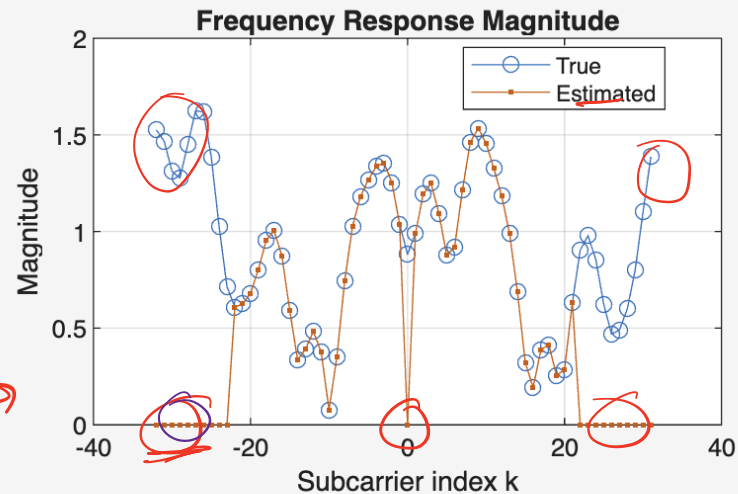
cross correlation
 $\times \text{corr}()$ $\rightarrow$ 64

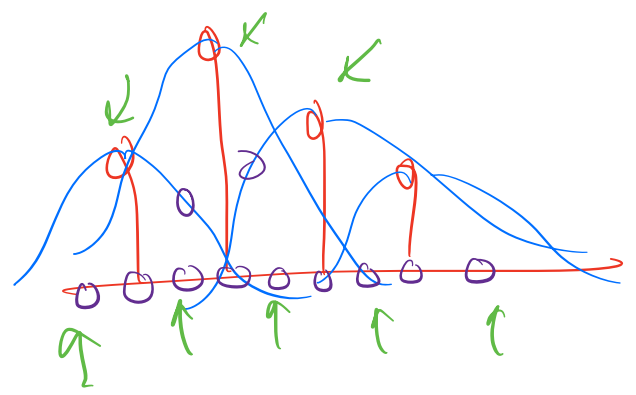
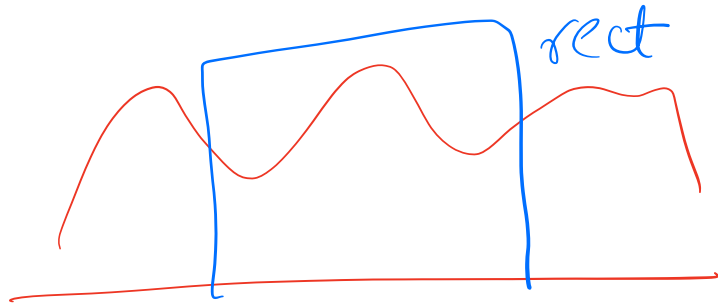


Channel Estimation (Lab 3)

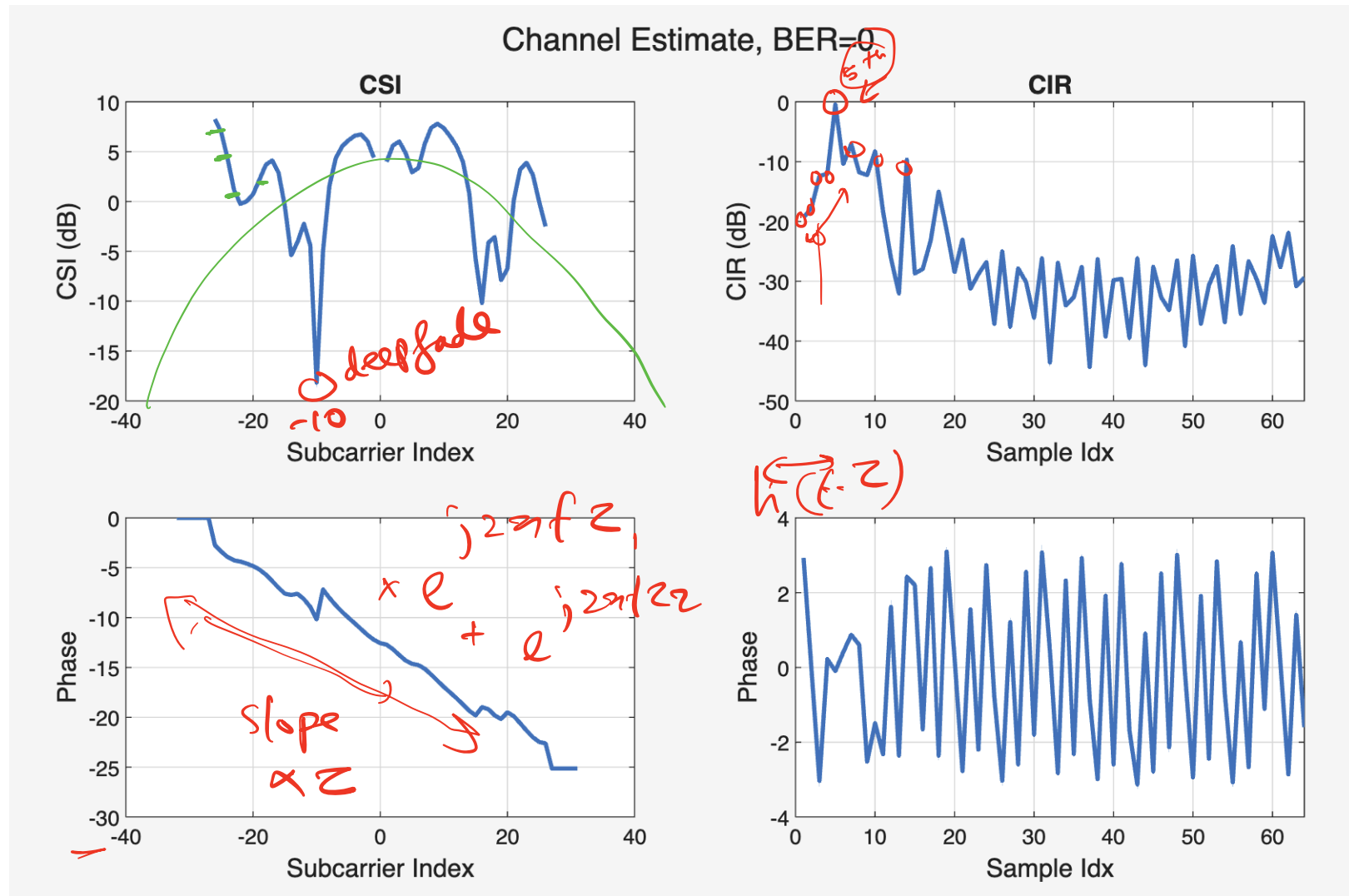


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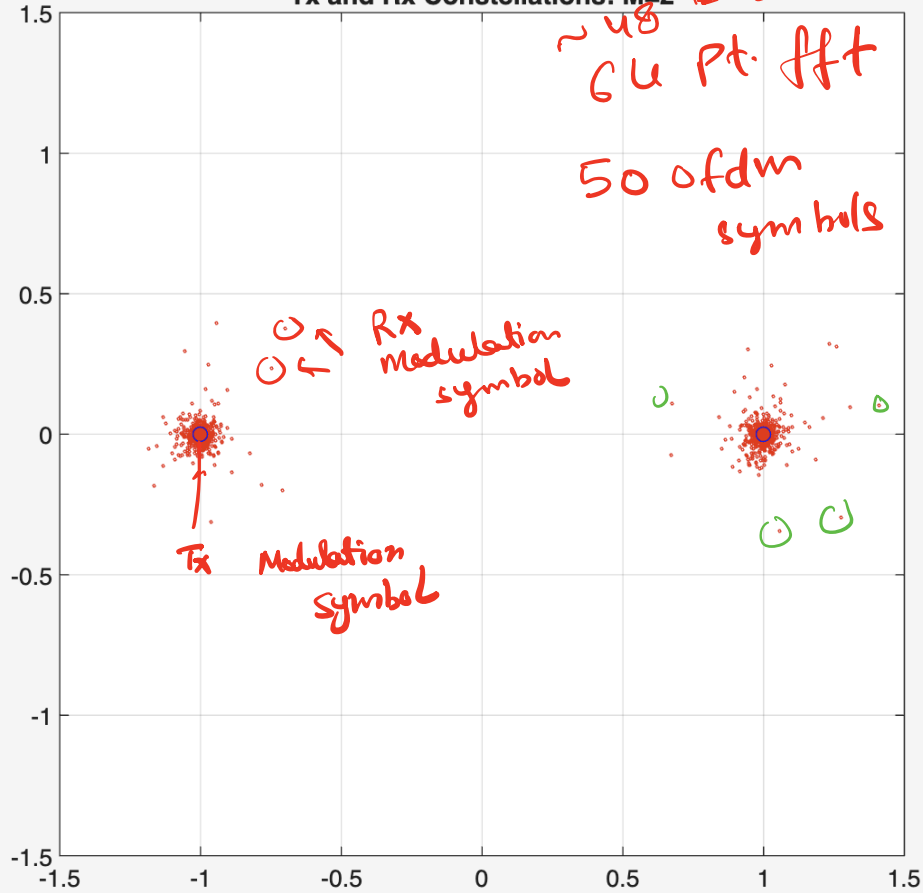


Channel Estimation (Lab 4 OFDM)



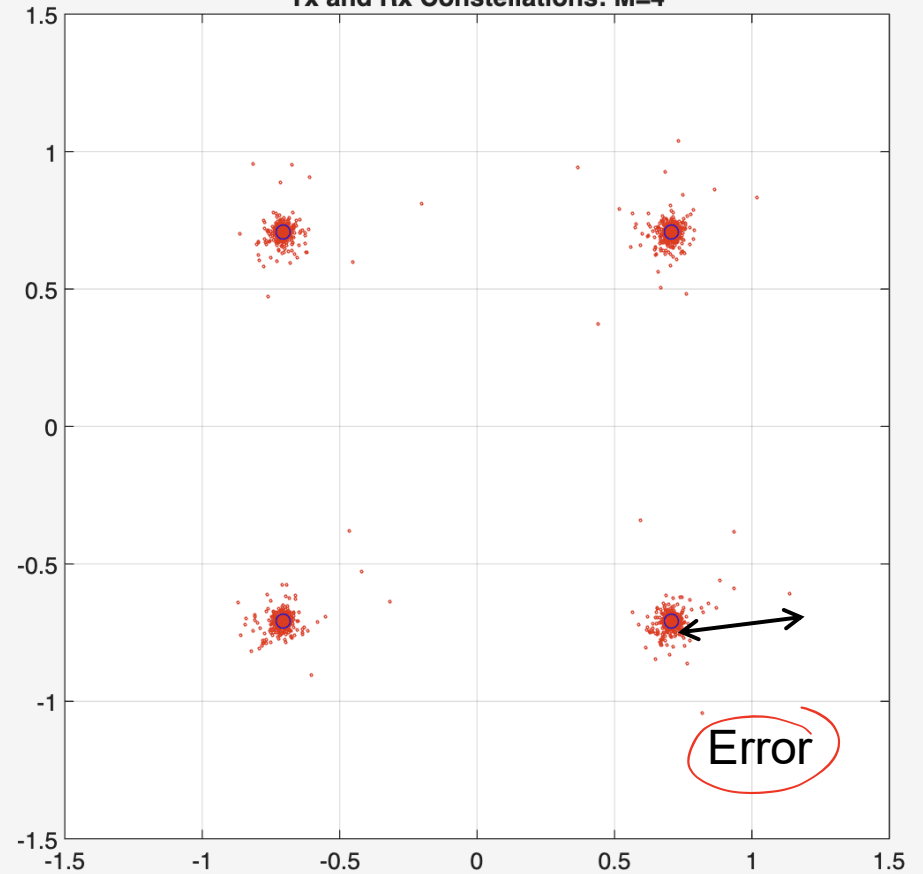
Constellation Diagram

Tx and Rx Constellations: M=2



BPSK

Tx and Rx Constellations: M=4

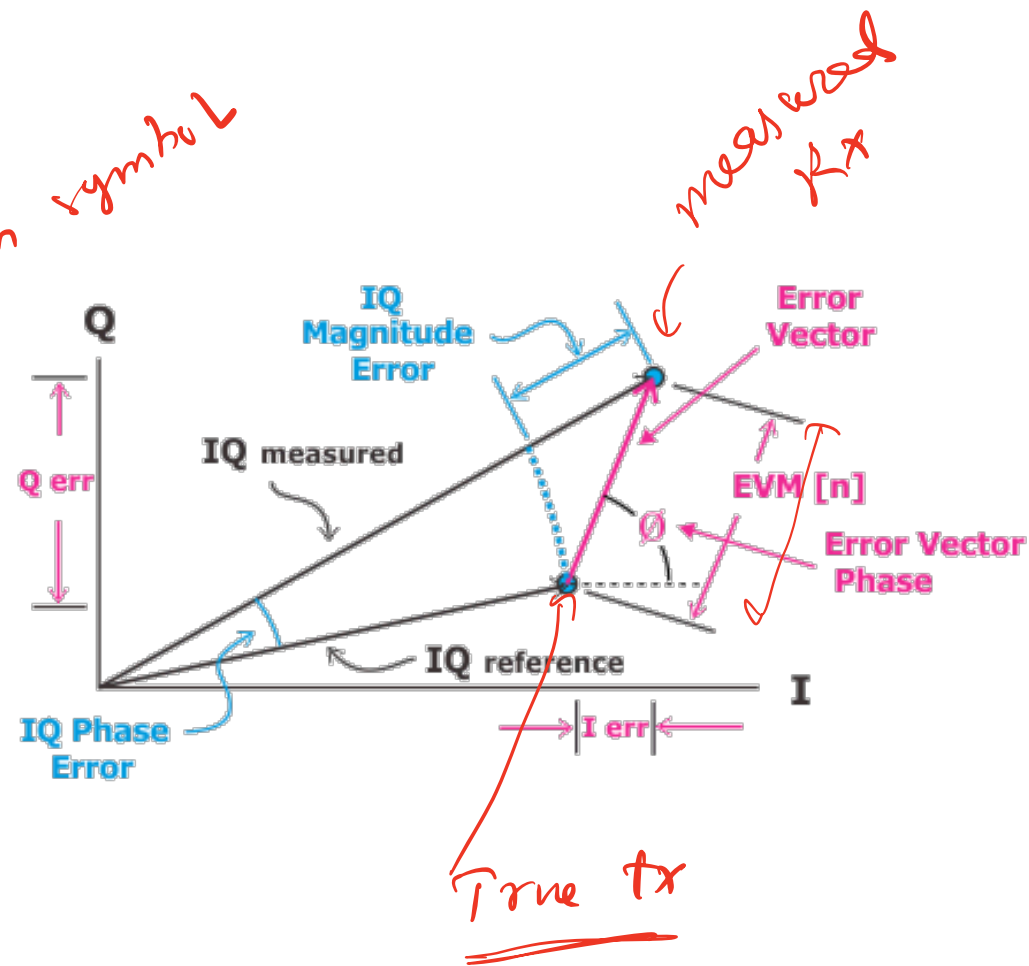


QPSK

Error Vector Magnitude

$$\% \text{ EVM} = \frac{\sqrt{\frac{1}{N} \sum_{n=0}^{N-1} \text{Ierr}[n]^2 + \text{Qerr}[n]^2}}{\text{EVM Normalization Reference}} \times 100\%$$

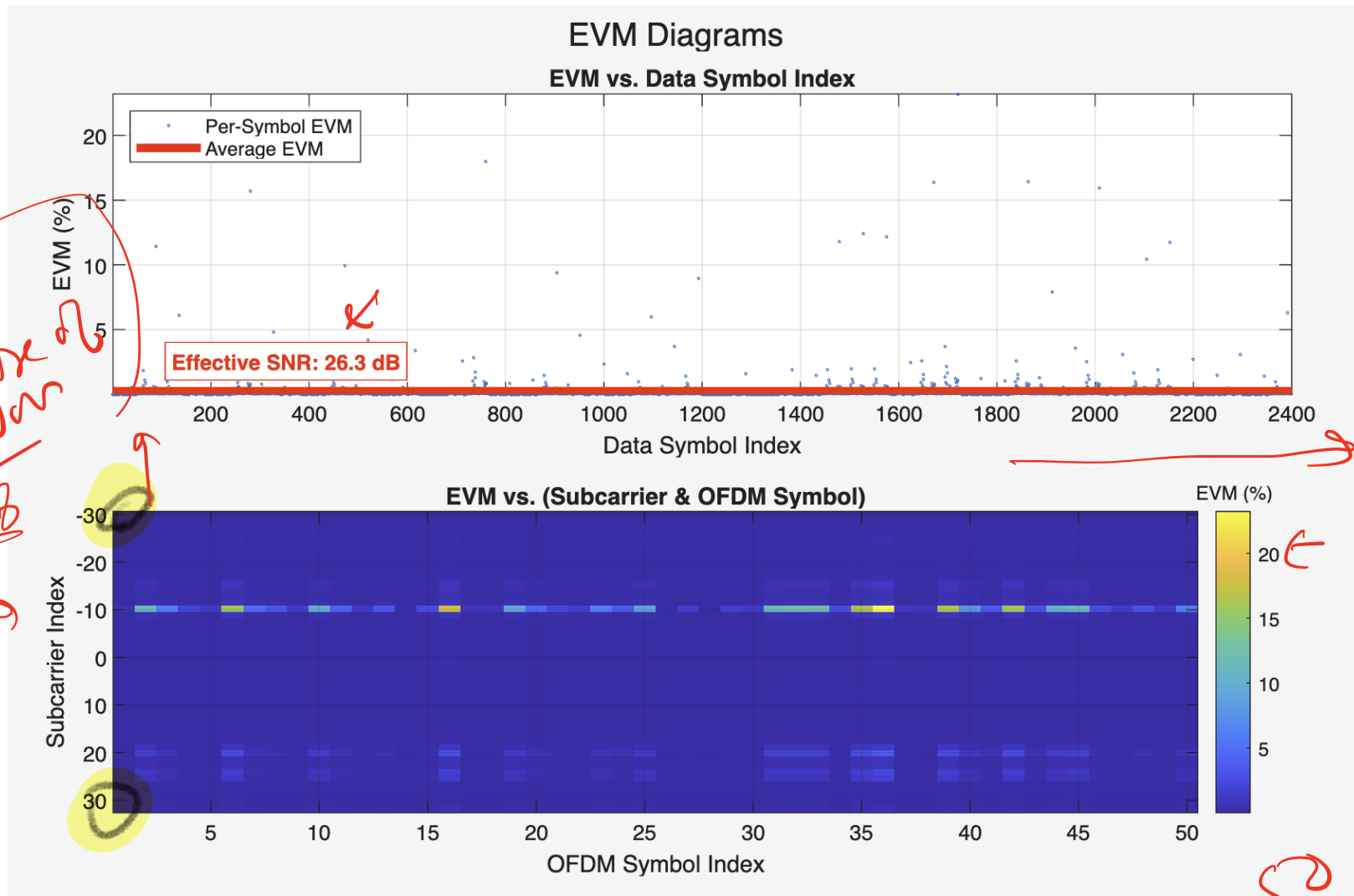
where:
 n = symbol index
 N = number of symbols
 Ierr = I Ref - I Meas
 Qerr = Q Ref - Q Meas



EVM Normalization Reference:

Constellation Maximum (default): sets the normalization value to the maximum constellation magnitude.

EVM (Error Vector Magnitude) Diagram



SNR is inverse of EVM

48

-10 sub carrier index

EVM is very high why?

48x10

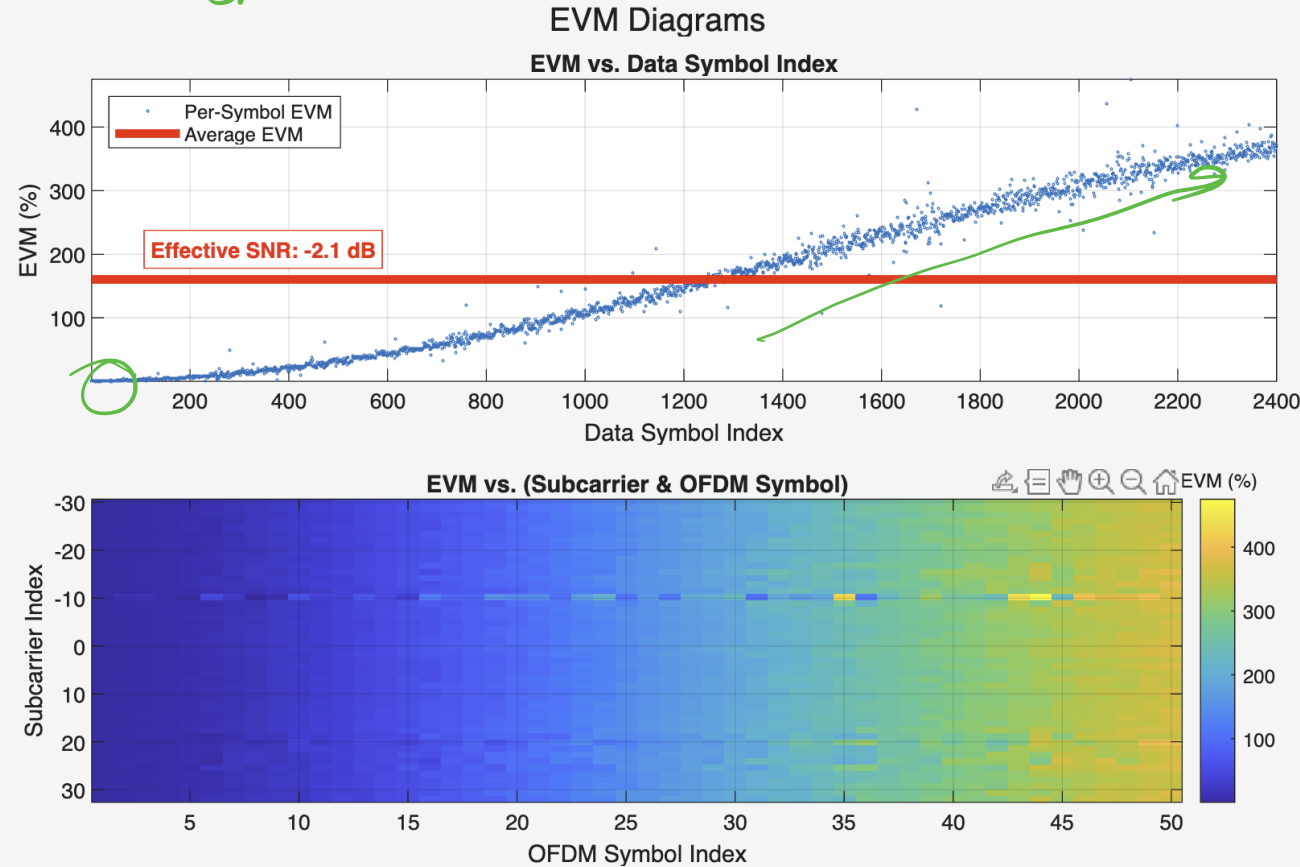
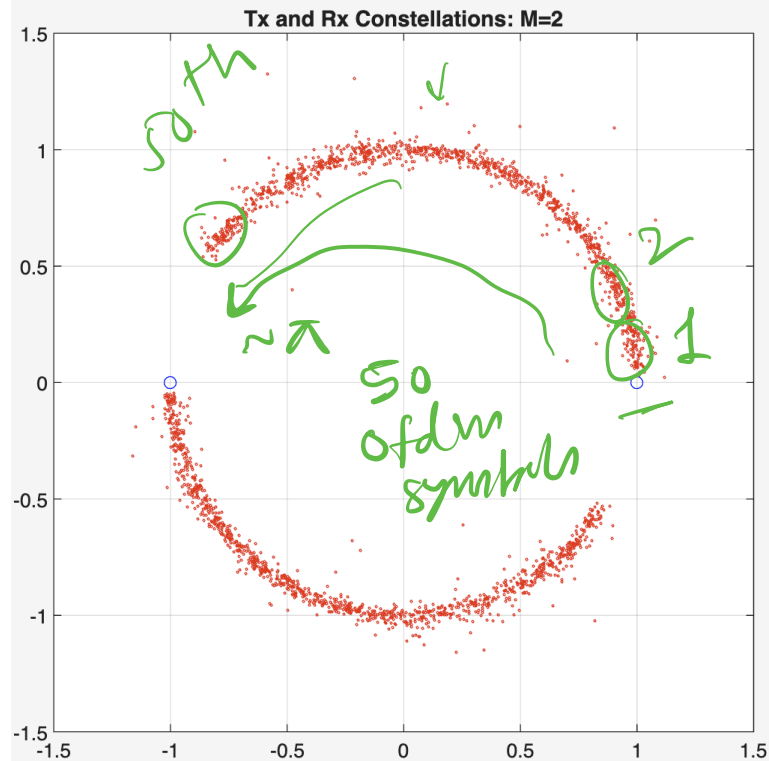
50

CFO Offset

$$x \rightarrow y = x e^{j 2\pi f_{\text{CFO}} t}$$

100 Hz 7
t ~ 100
ms

- Add CFO
- Don't correct it yet

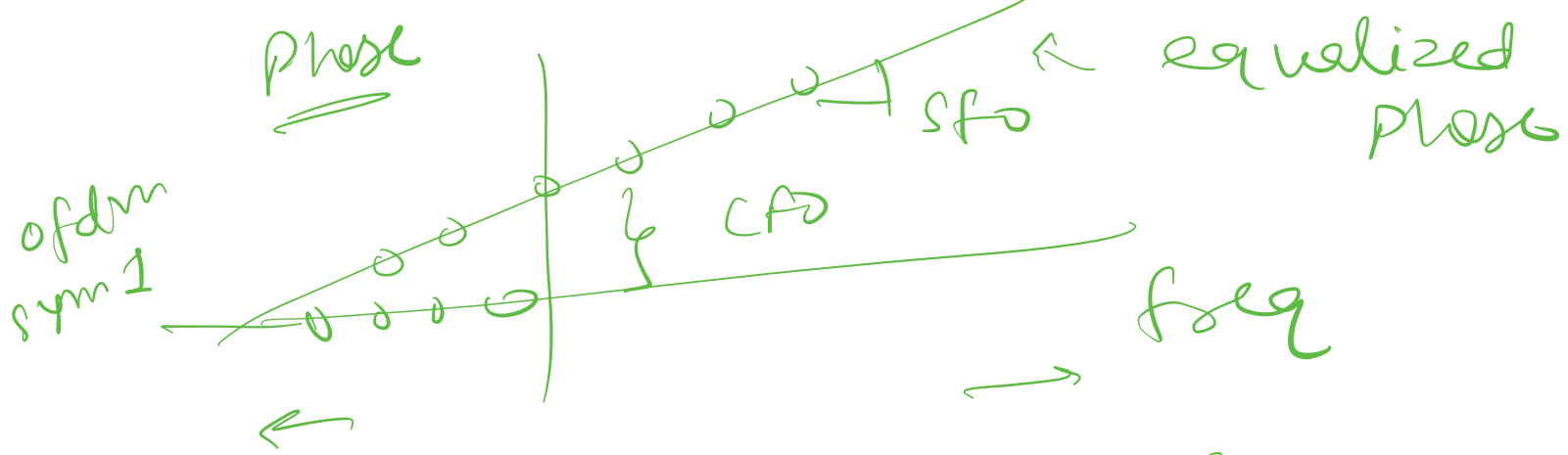


$$64 + 16 = 80 \text{ samples}$$

$$20 \text{ MHz} = \frac{1}{20} \cdot 0.05 \text{ ms / symbol}$$

$$80 \times 0.05 = 4 \text{ ms / ofdm symbol}$$

$$4 \times 50 = 200 \text{ ms / packet}$$



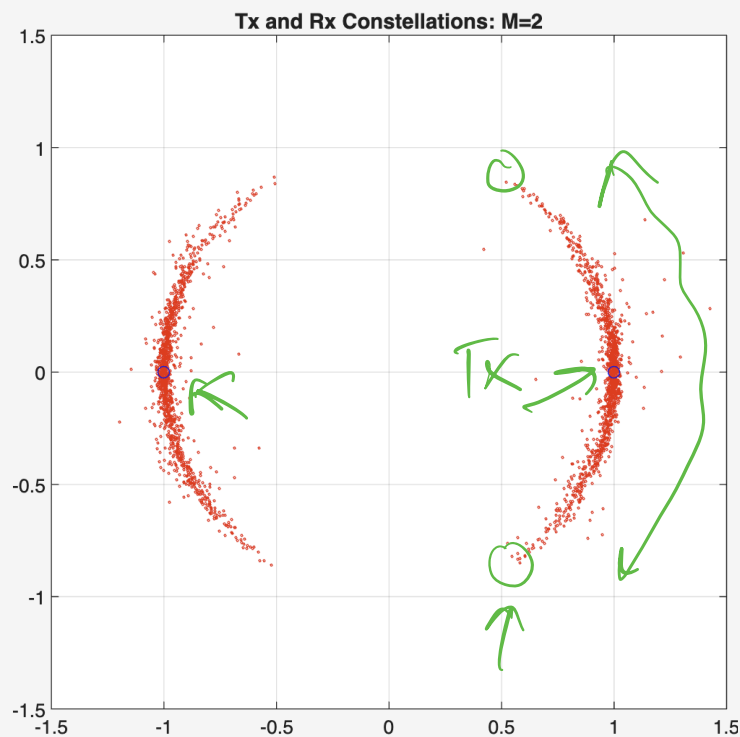
ofdm sym 50

SFO slope is high

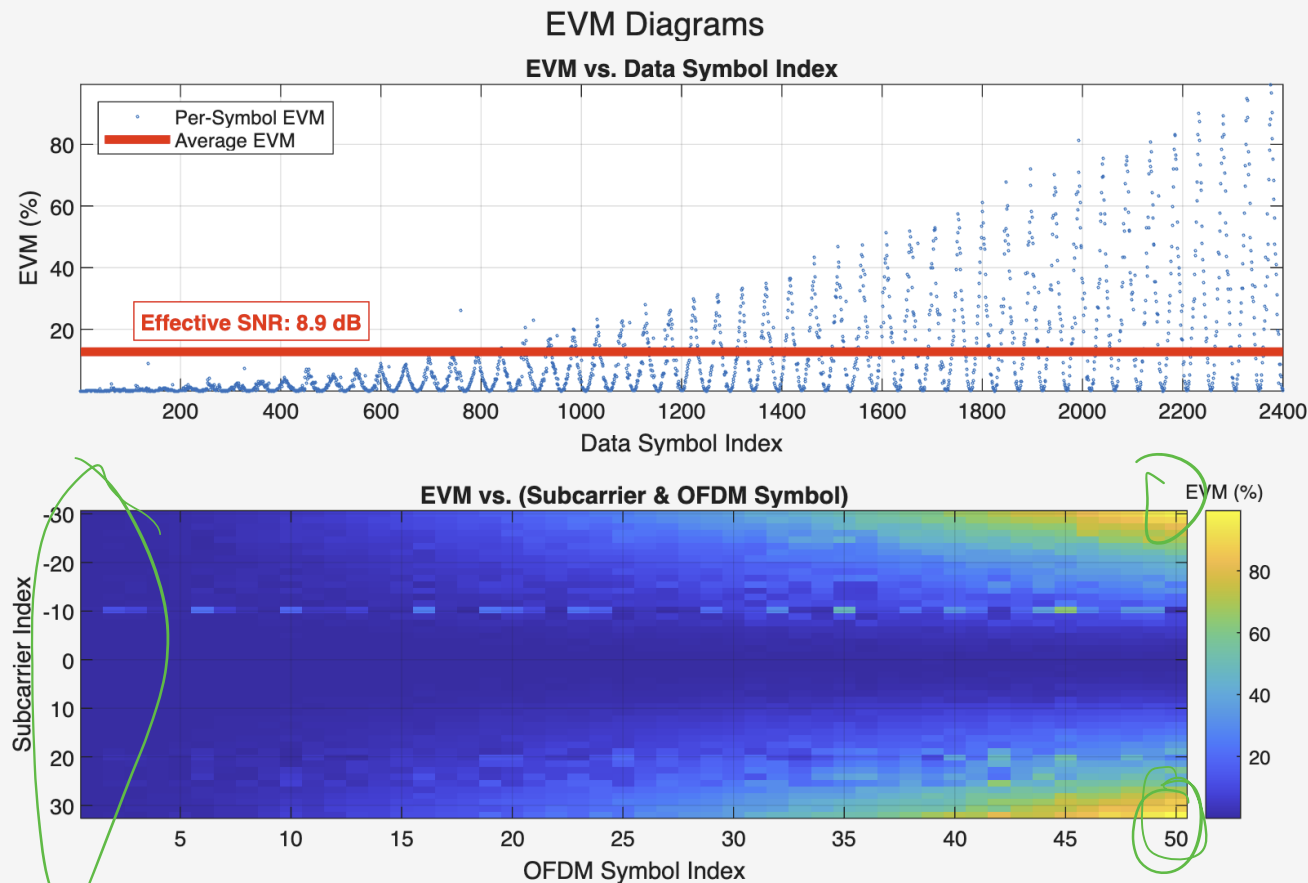
only SFO Offset

NO CFO

no phase correction (either CFO or SFO)



58th packet
corner subcarriers



Tx samp rate $\uparrow$ 100Hz
 $f_s = 20 \text{ MHz}$
Rx samp rate $f_r = f_s + \text{SFO}$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
 - Residual SFO (Sampling Frequency Offset)
- 

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
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When we sample the signal there is a residual sampling offset: $n\delta T_s$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

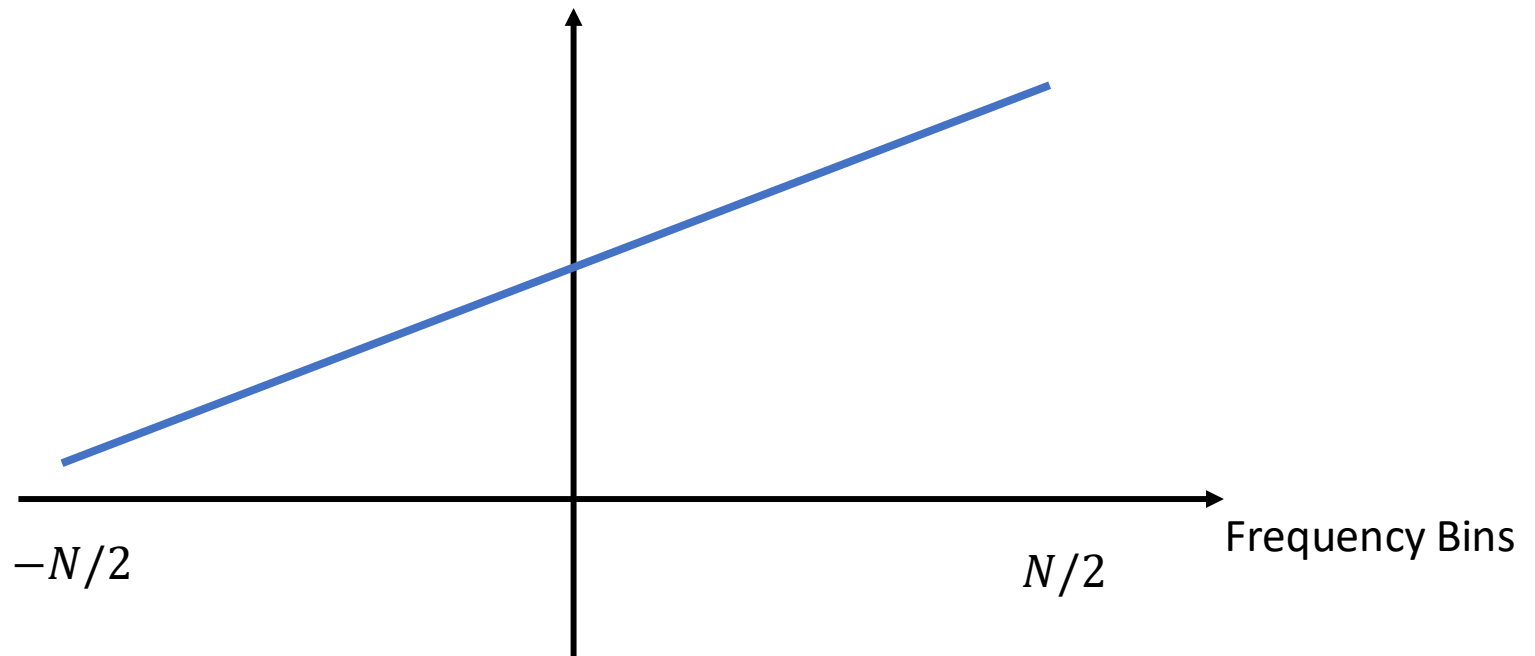
$$Y_1[f] = X_1[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

$$Y_2[f] = X_2[f] e^{j \frac{2\pi f (n+N+CP) \delta T_s}{N} - 2\pi \delta f_c (n+N+CP) T_s}$$

Phase accumulation: $\Delta\phi = 2\pi f \frac{(N+CP)\delta T_s}{N} - 2\pi \delta f_c (N+CP) T_s$

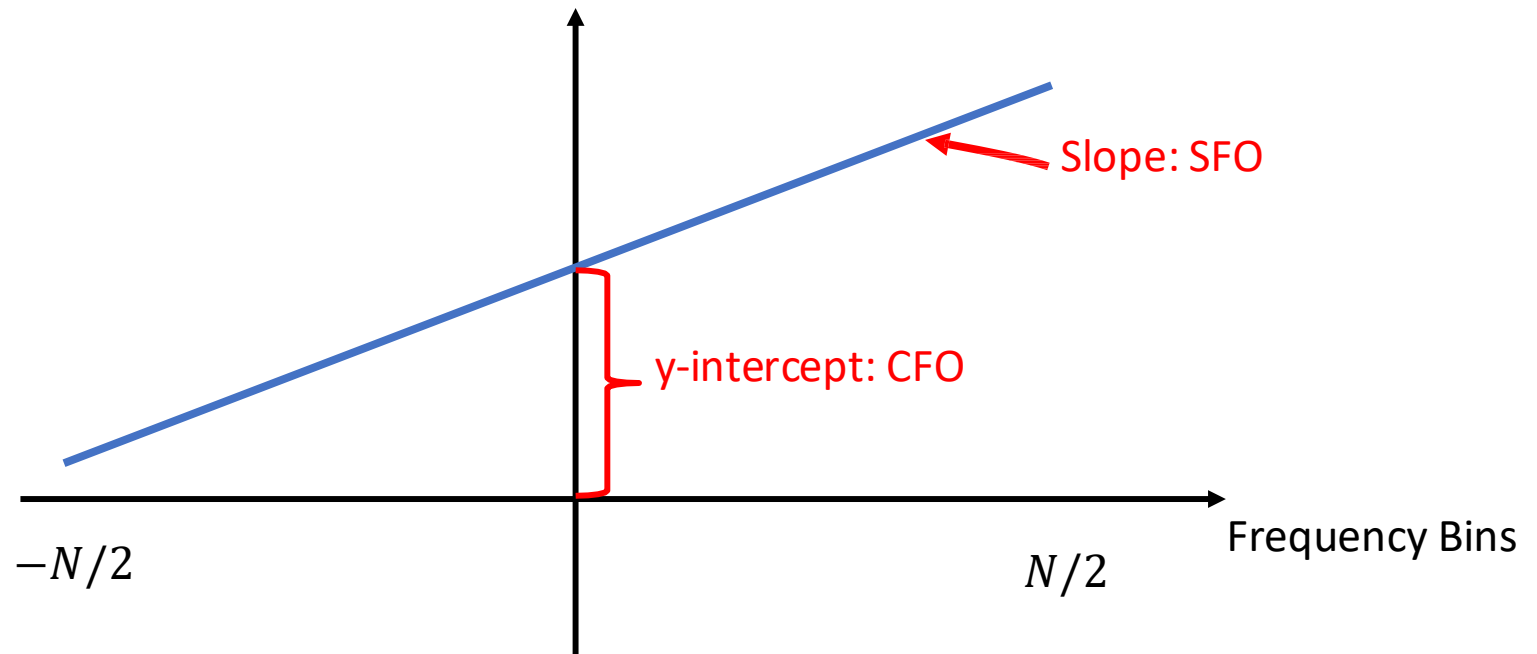
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
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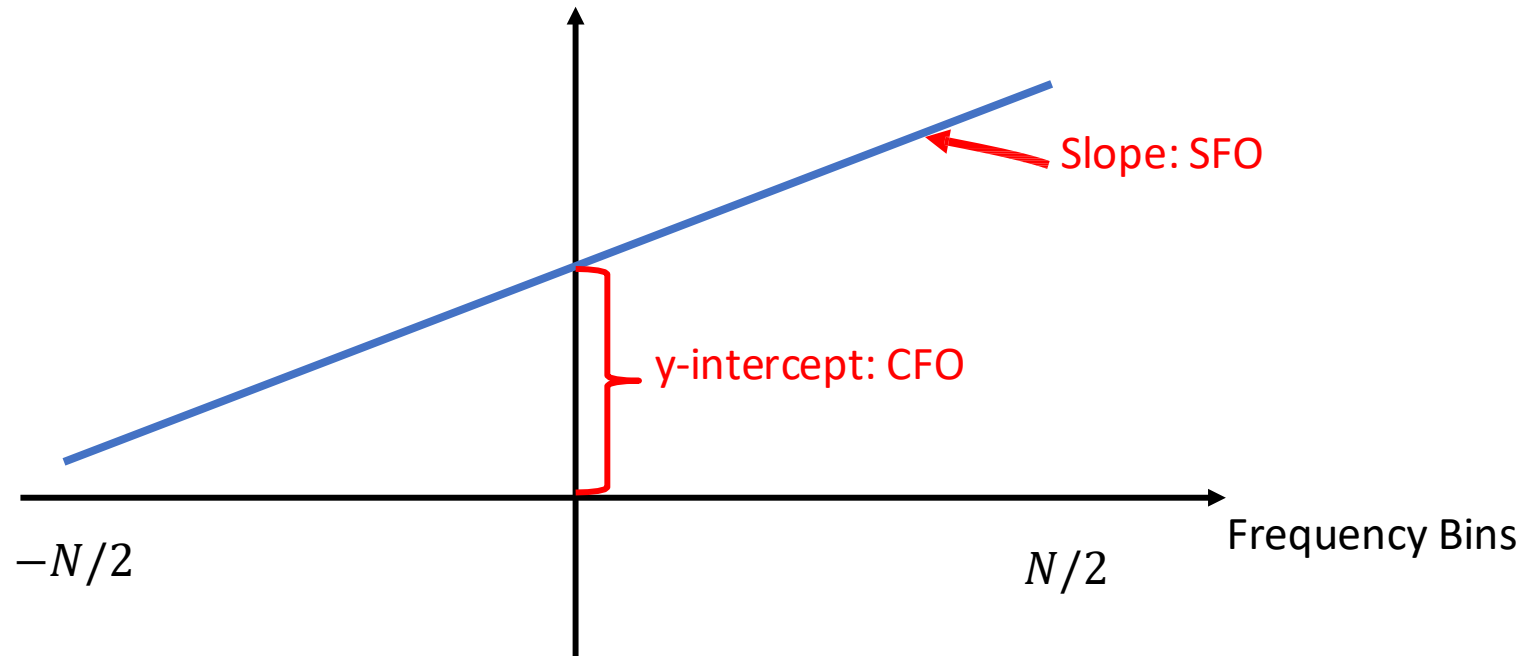
Phase Tracking

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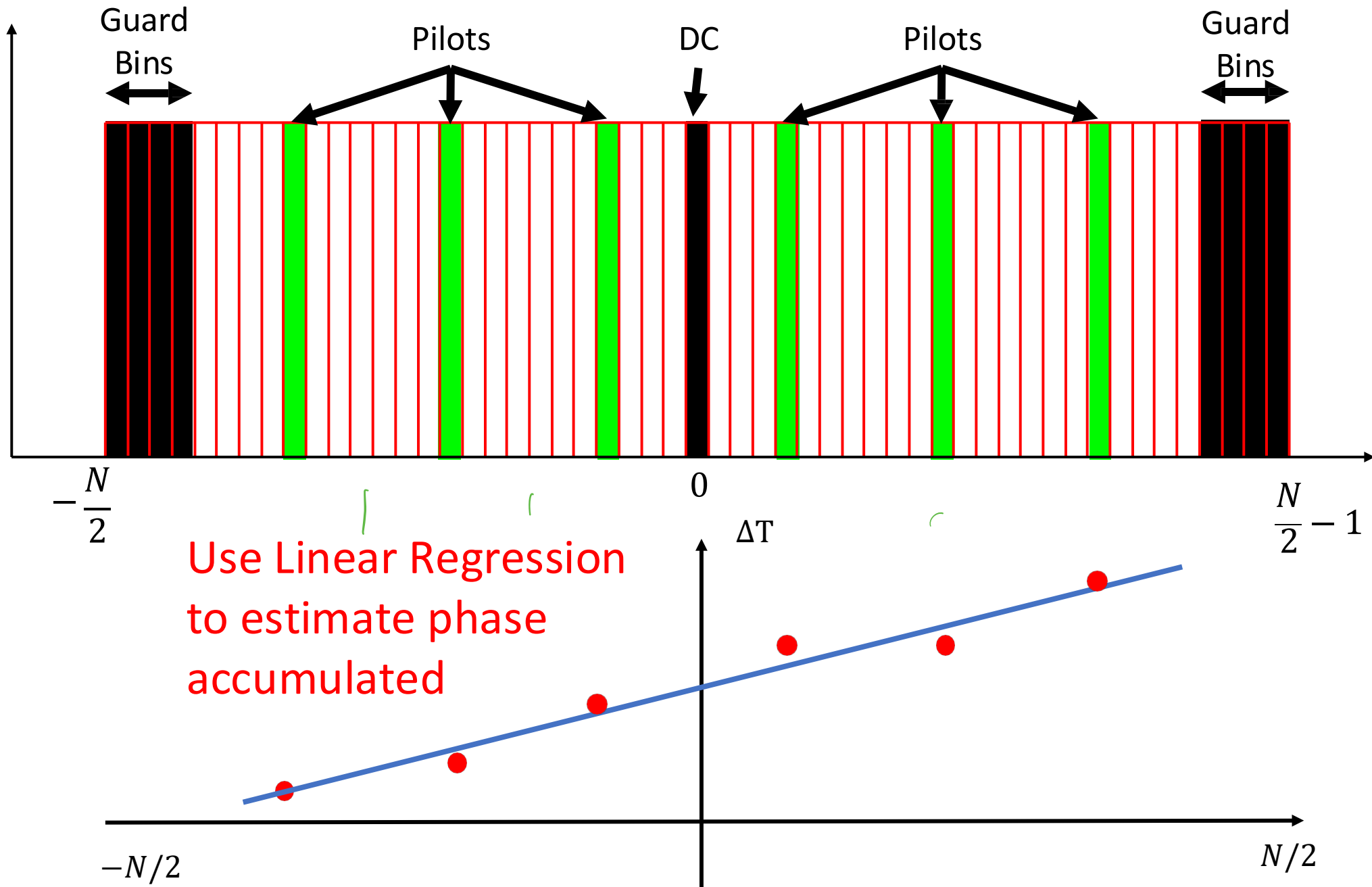
OFDM Phase Tracking

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



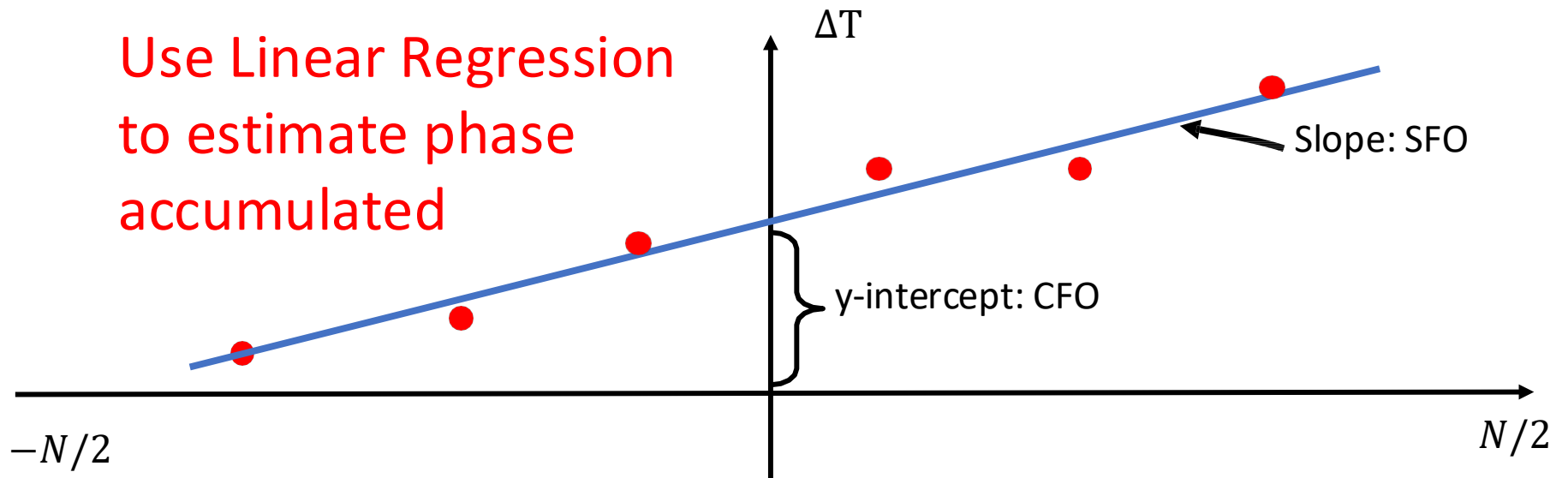
- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

OFDM Phase Tracking



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

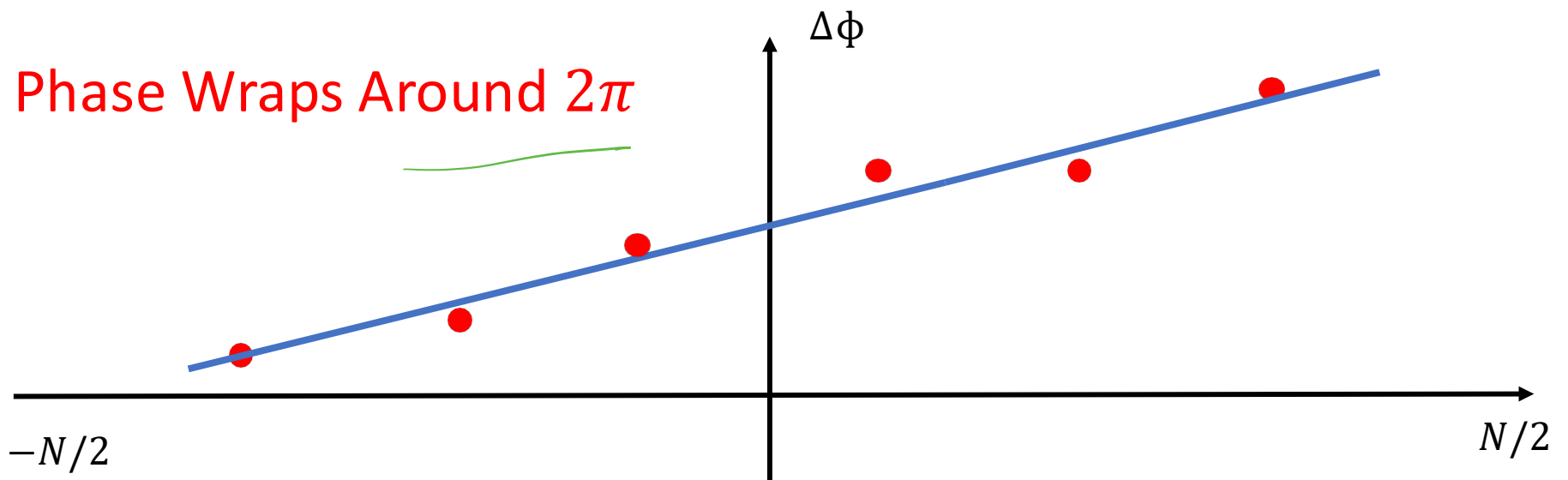


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

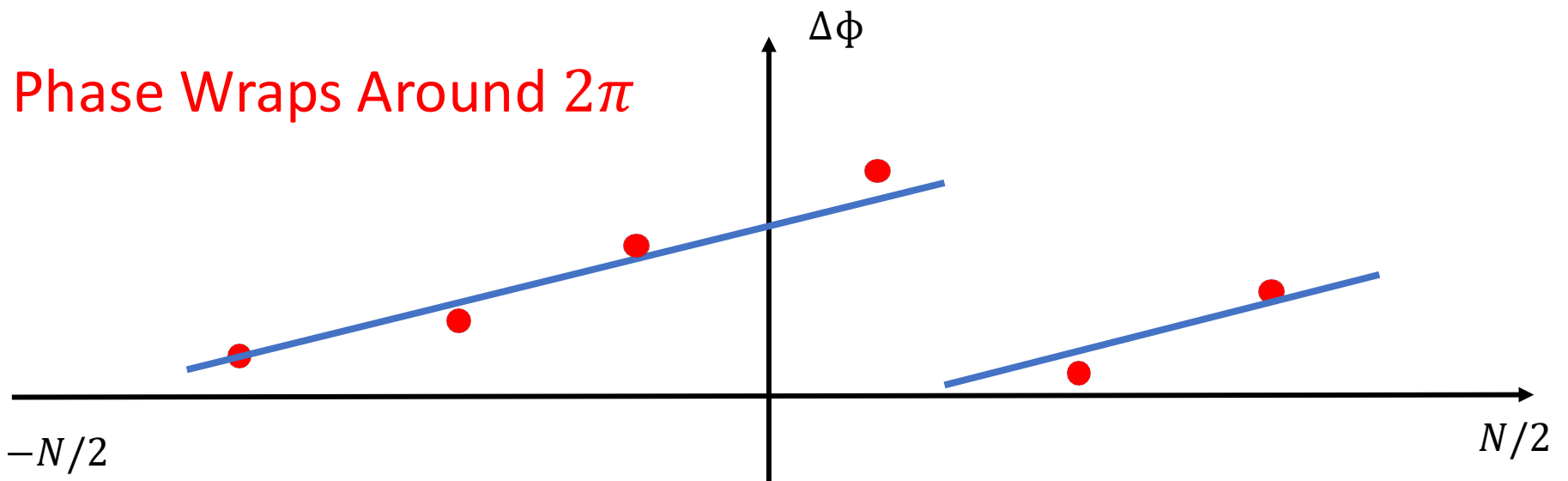
$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

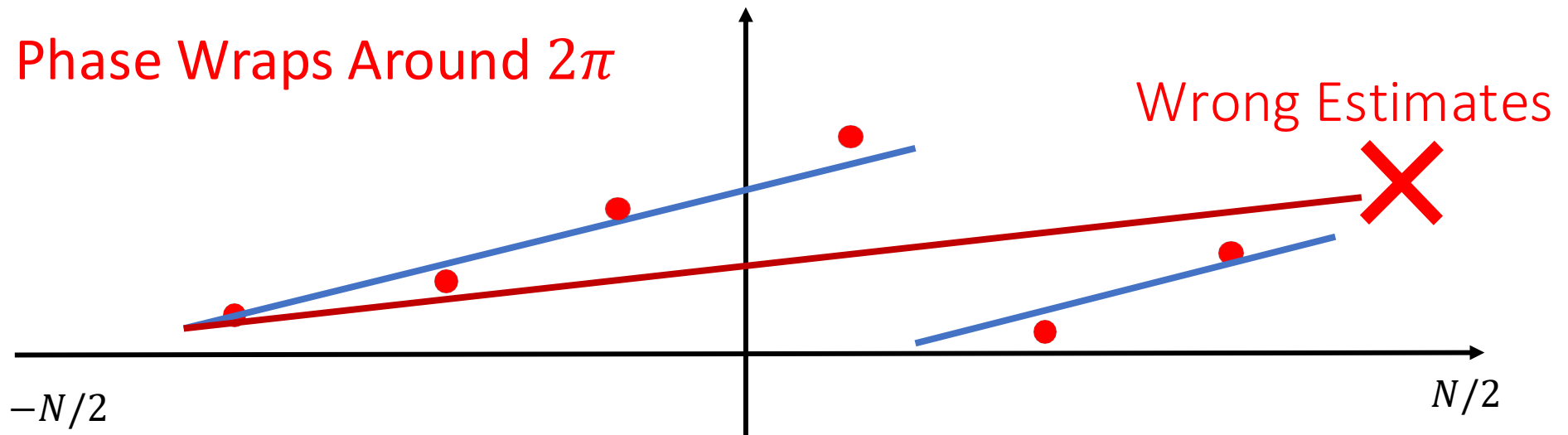
Be Careful!



Be Careful!



Be Careful!

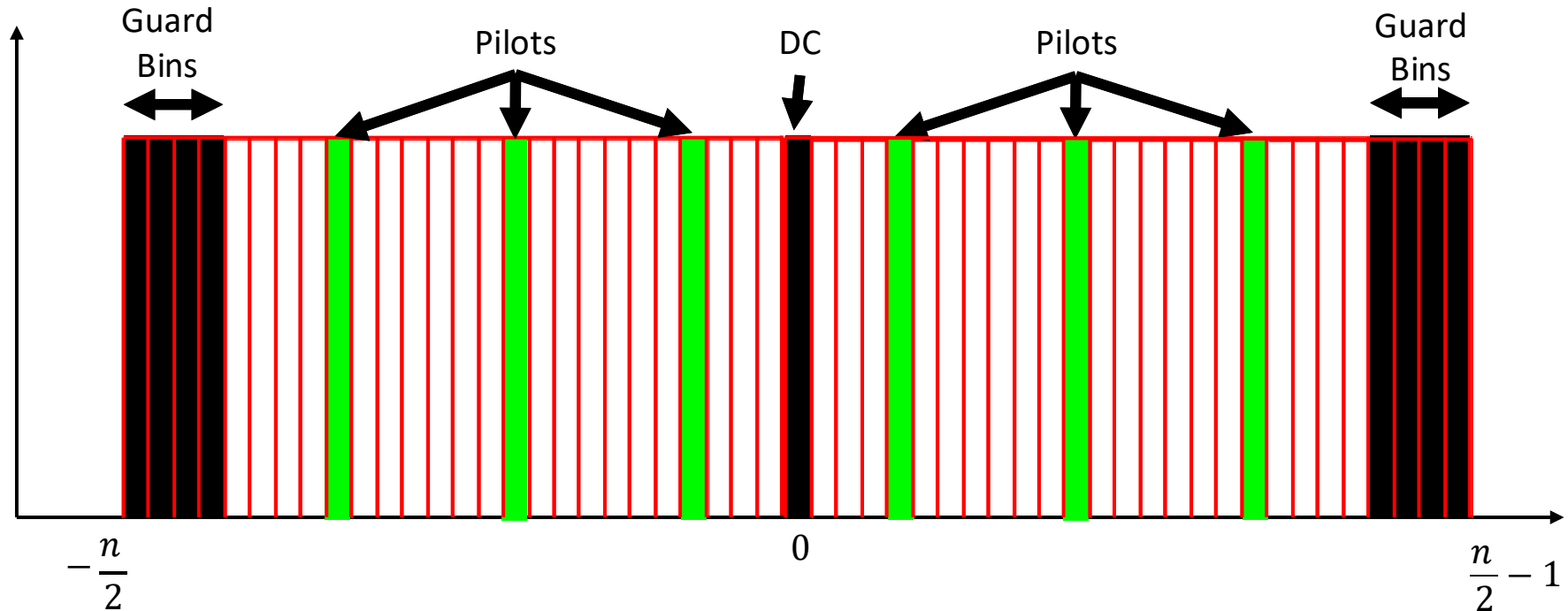


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

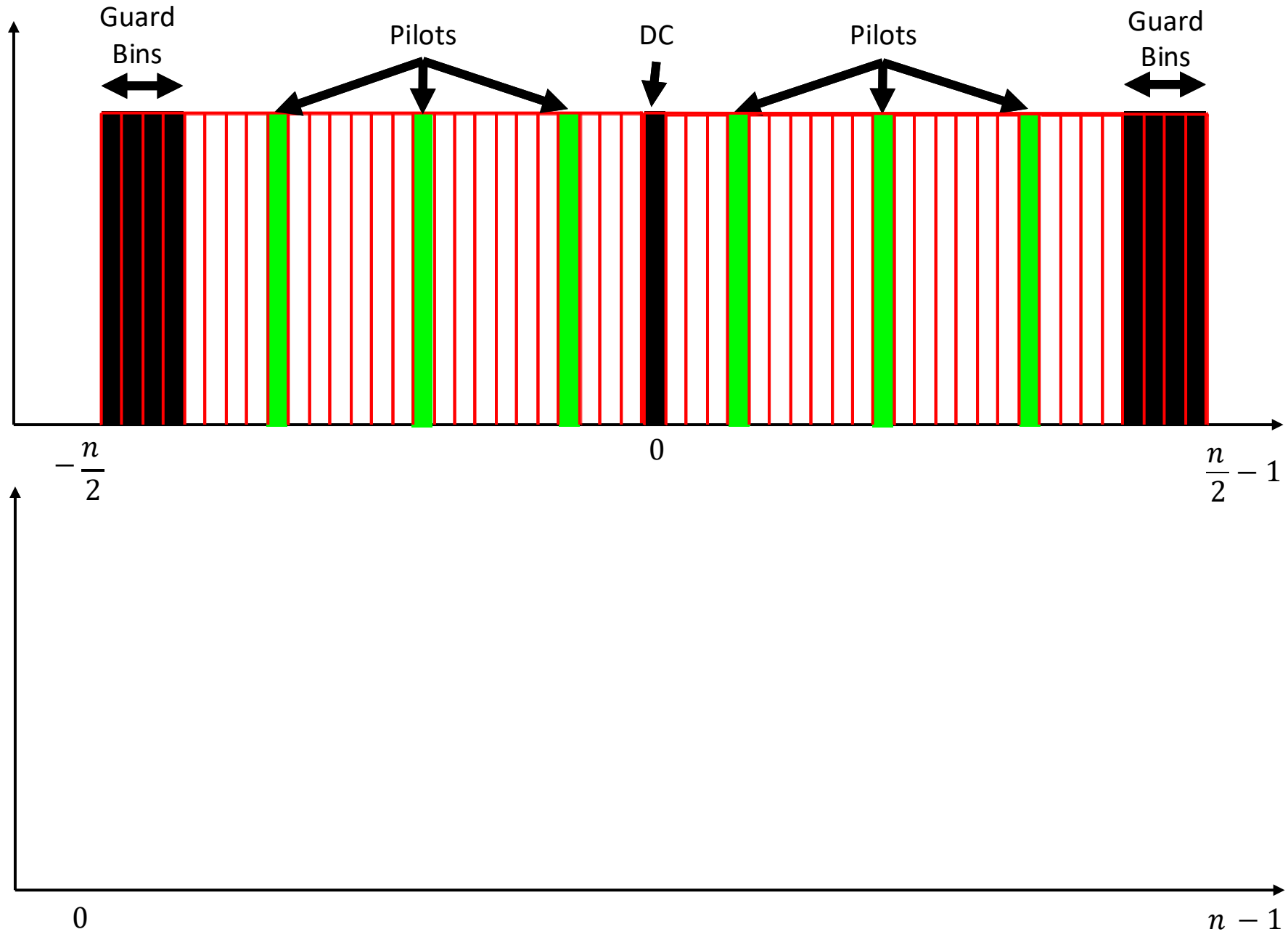
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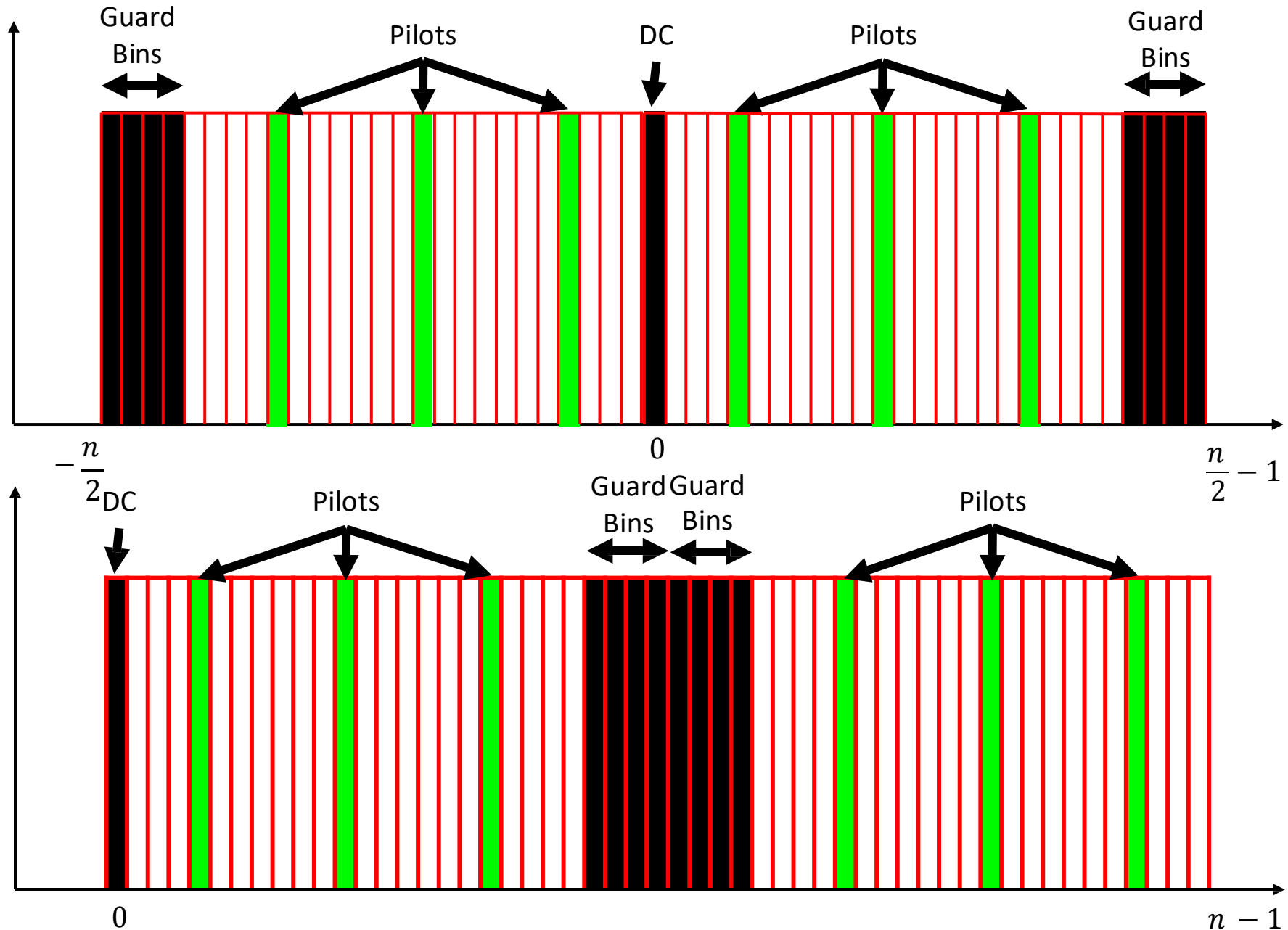
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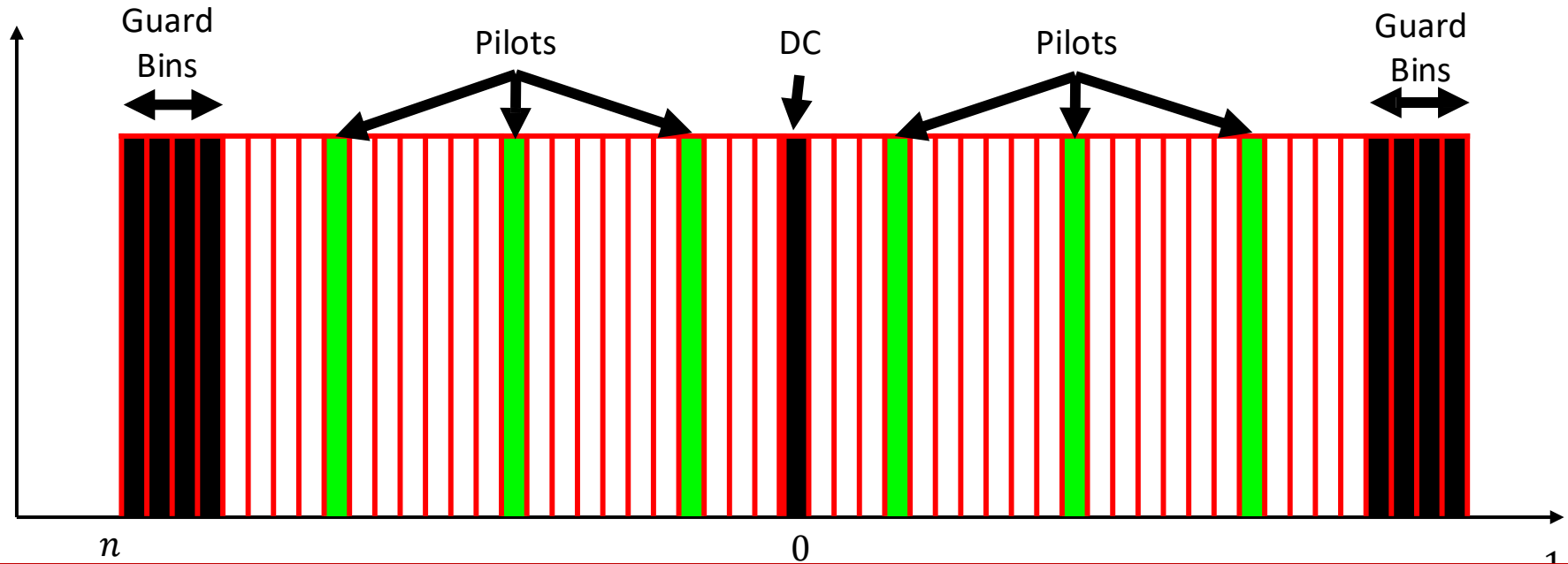
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$

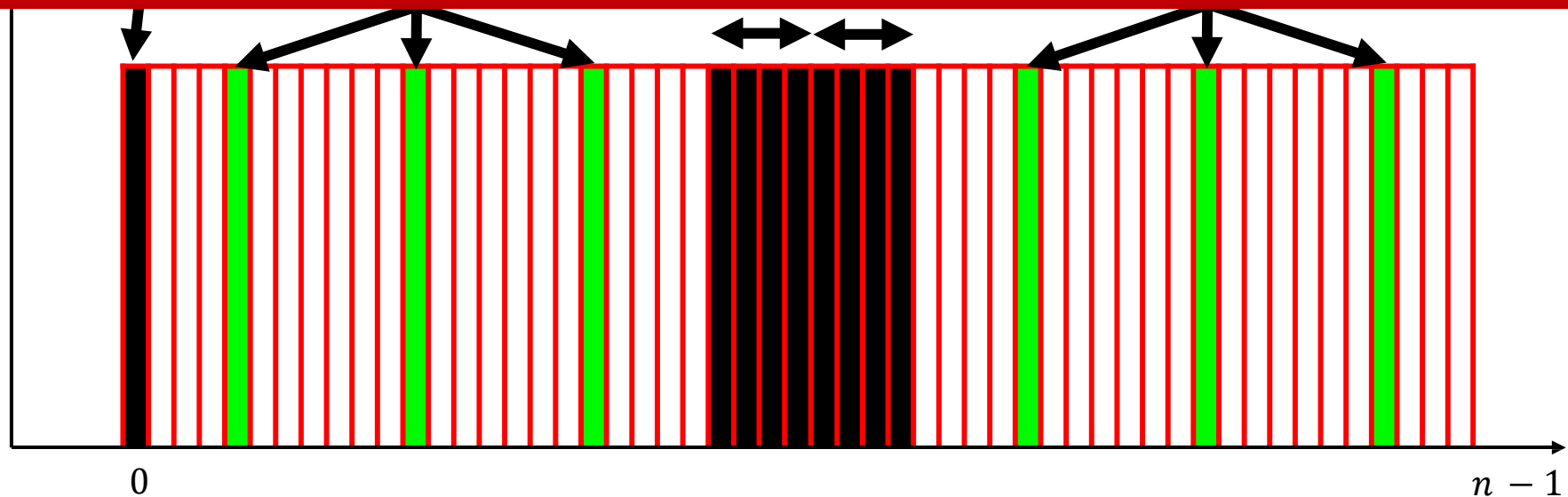


Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$



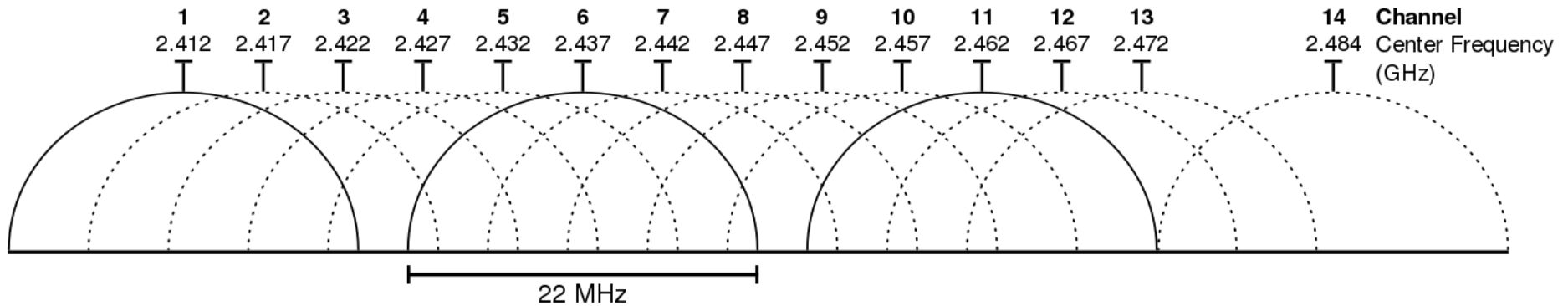
Need to `fftshift()` function in MATLAB + provided functions to convert indexing.



Today's Class

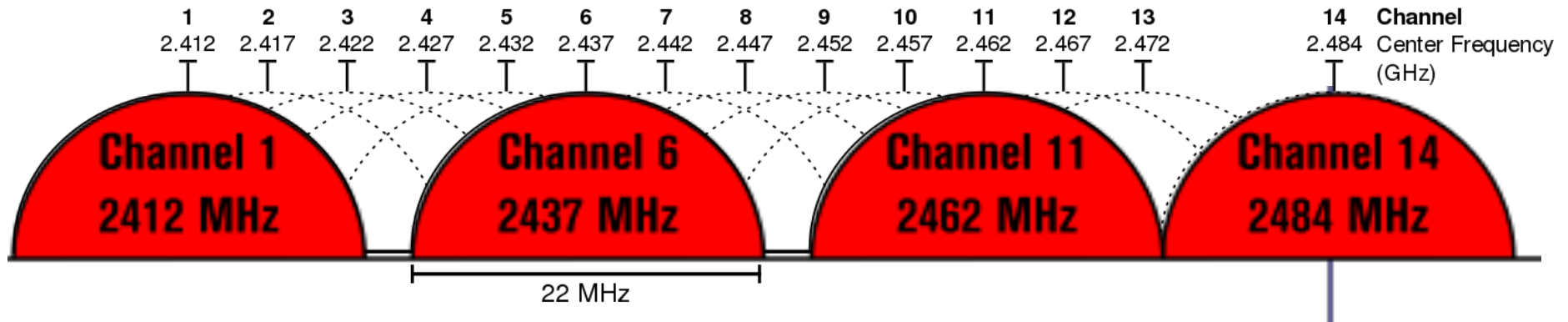
- OFDM Revision
 - HW/Lab 3, 4
 - SFO and Residual Phase correction (ECSE6560 topic)
- **MIMO Intro**
 - **MIMO in WiFi Standards**
 - **MIMO Intro**
 - **MIMO vector representation**

WiFi Standards from WiFi 1 to WiFi 6



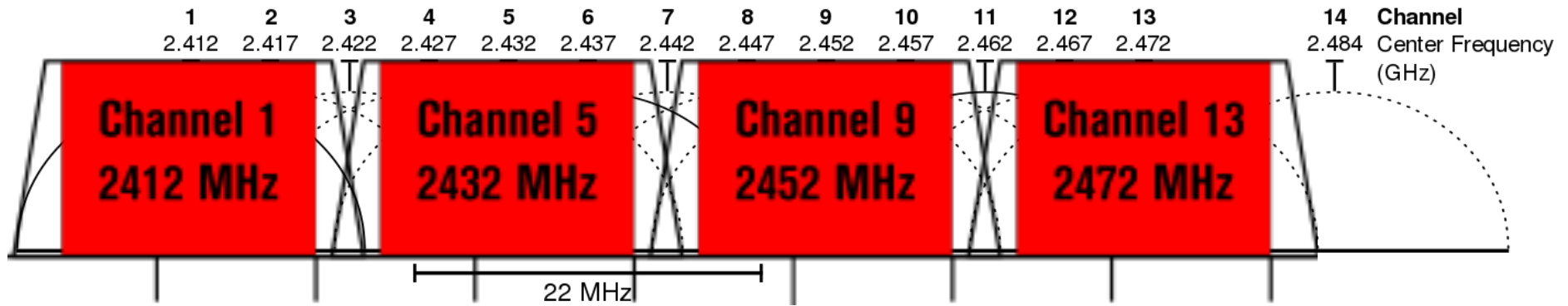
Version	Year	Technology	Modulation	Freq.	Bandwidth	Maximum Data Rate
WiFi 1 (802.11b)	1999	DSSS	DBPSK, DQPSK,	2.4 GHz	22 MHz	11 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



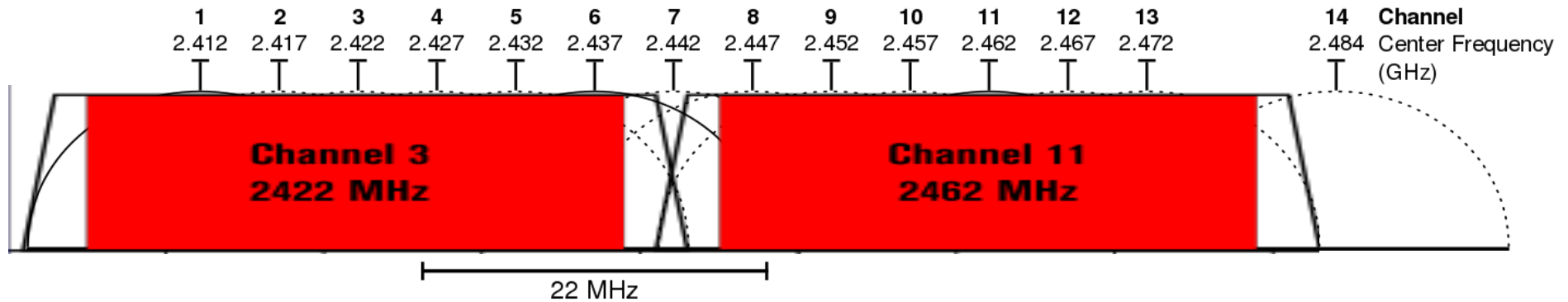
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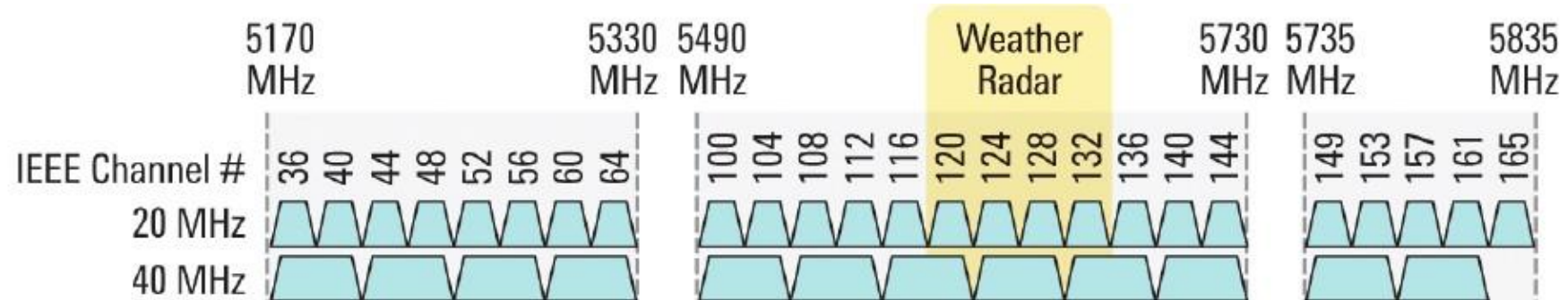
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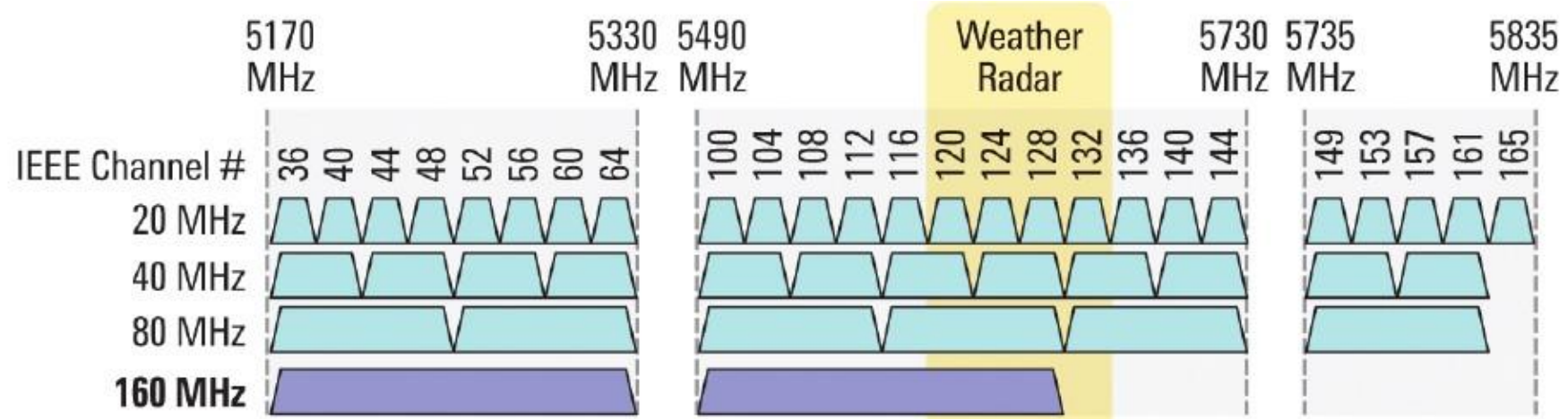
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WiFi 4 (802.11n)	2009	OFDM (N=64) MIMO (4x4)	BPSK, QPSK, 16-QAM, 64-QAM	2.4 GHz 5 GHz	20 MHz 40 MHz	600 Mb/s

WiFi Standards from WiFi 1 to WiFi 6



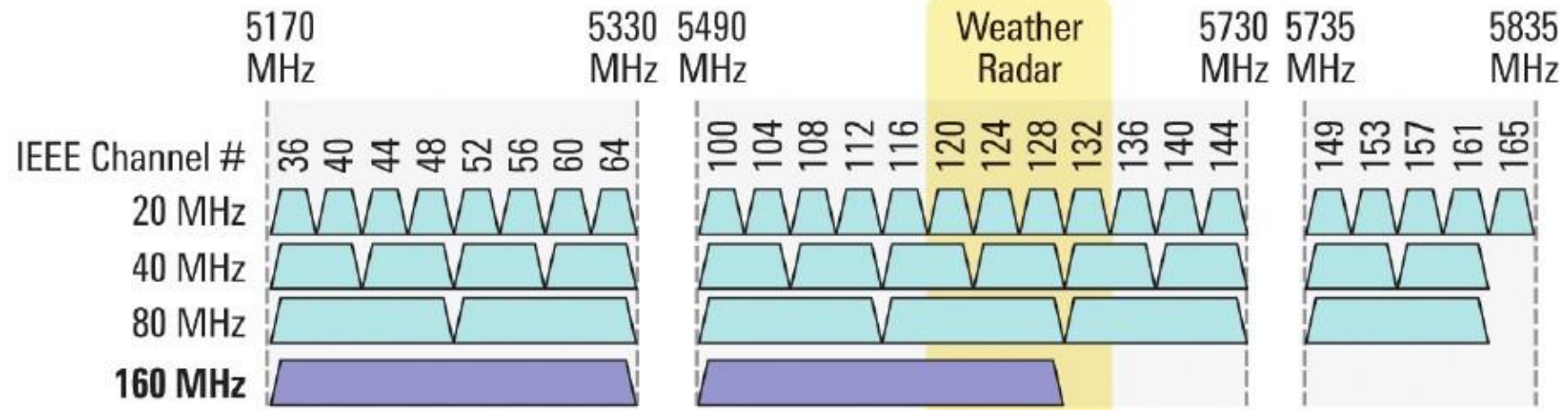
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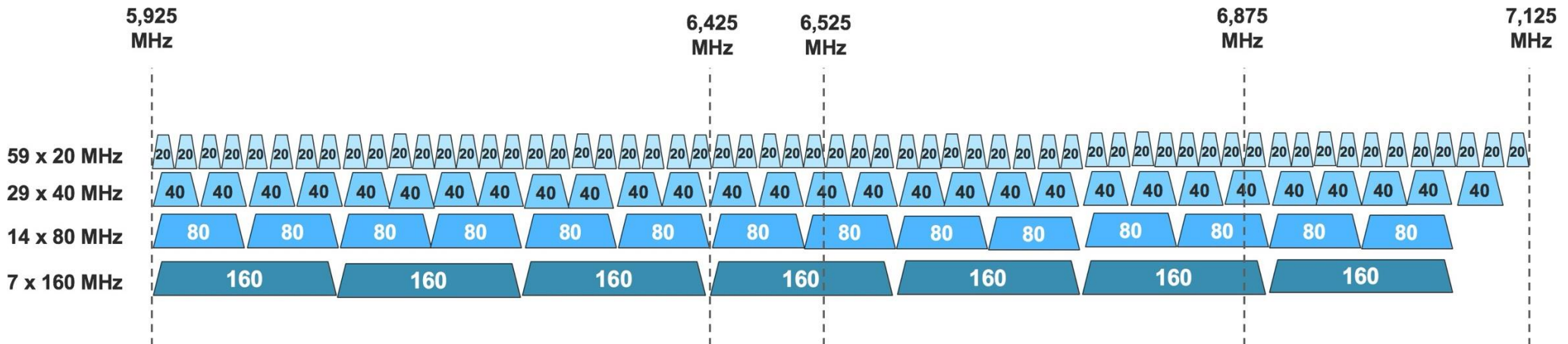
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WiFi 5 (802.11ac)	2014	OFDM (N=64, 128, 256, 512) MIMO (8x8) MU-MIMO (Downlink)	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM	5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	6.933 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



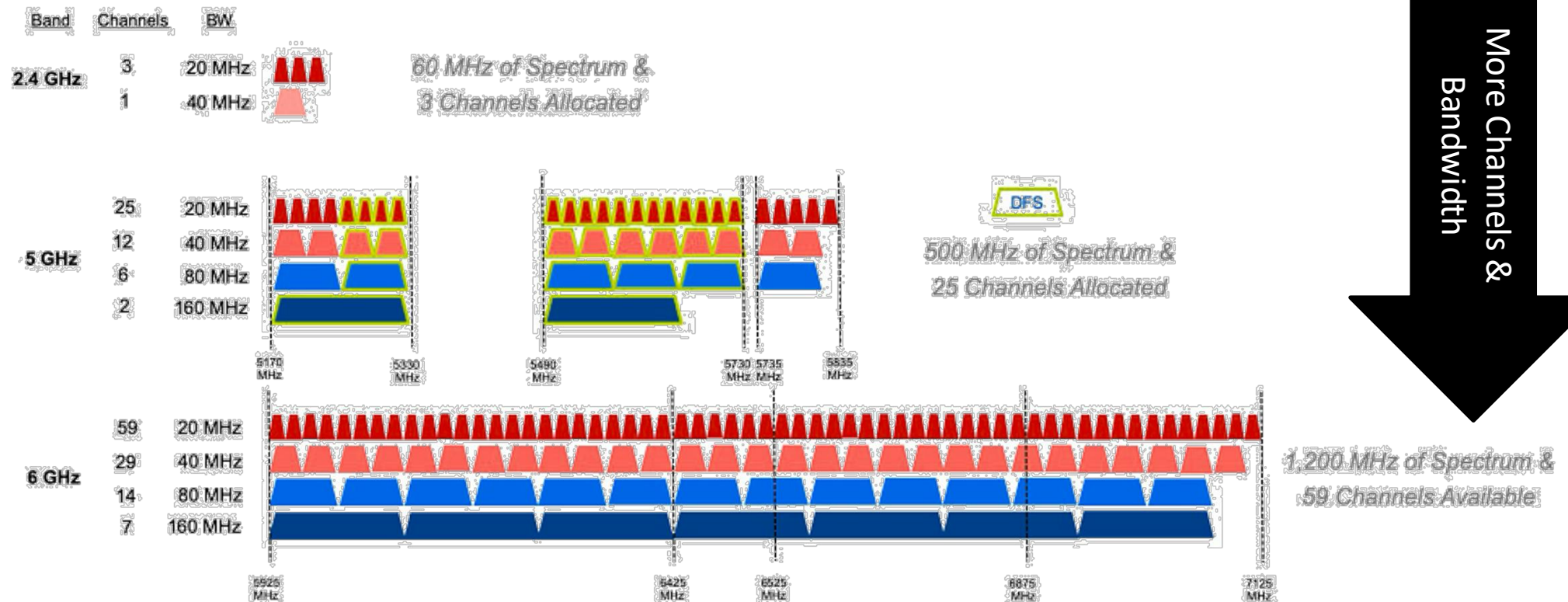
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WiFi 6 (802.11ax)	2019	OFDM (N=256, 512, 1024, 2048) MIMO (8x8) MU-MIMO (Up & Down) OFDMA	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	9.608 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



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WiFi 6 (802.11ax) WiFi 6E	2020	OFDM (N=256, 512, 1024, 2048) MIMO (8x8) MU-MIMO (Up & Down) OFDMA	BPSK, QPSK, 16-QAM, 64-QAM 256-QAM, 1024-QAM	2.4 GHz 5 GHz 6 GHz	20 MHz 40 MHz 80 MHz 160 MHz	9.608 Gb/s

WiFi Standards from WiFi 1 to WiFi 6



WiFi 1

More MIMO Antennas

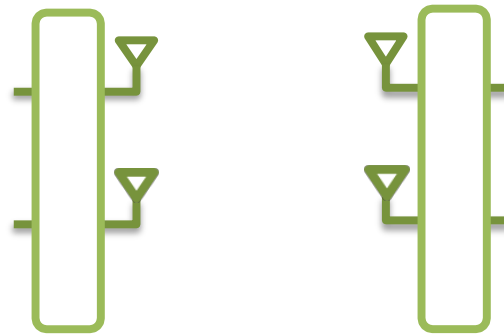
WiFi 5/6

MIMO: Multiple Input Multiple Output

So far: single input single output



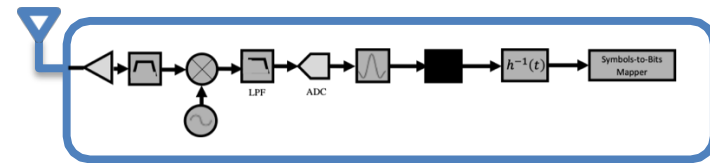
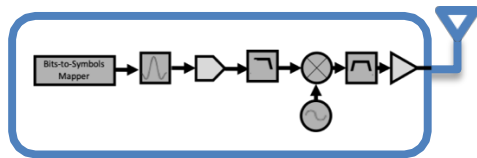
MIMO: multiple input multiple output



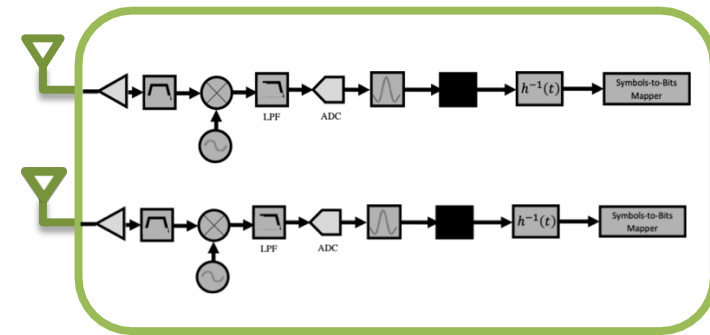
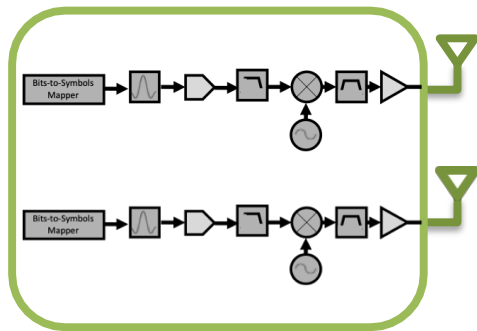
Increase capacity of channel using multiple transmit and receive antennas.

MIMO: Multiple Input Multiple Output

So far: single input single output

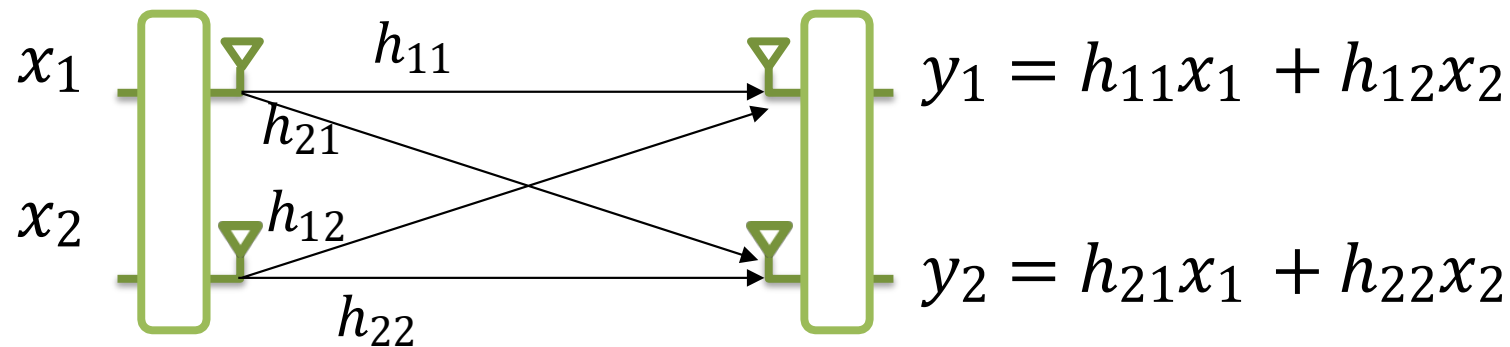


MIMO: multiple input multiple output



Increase capacity of channel using multiple transmit and receive antennas.

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

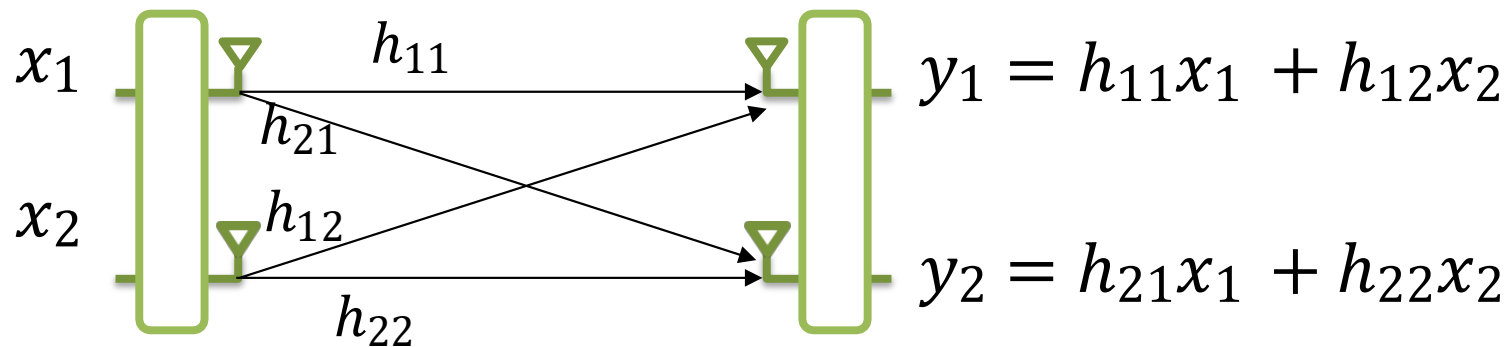
How to recover x_1 and x_2 ?

Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Transmit 2 packets at the same time!

MIMO: Multiple TX-RX streams



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

How to recover x_1 and x_2 ?

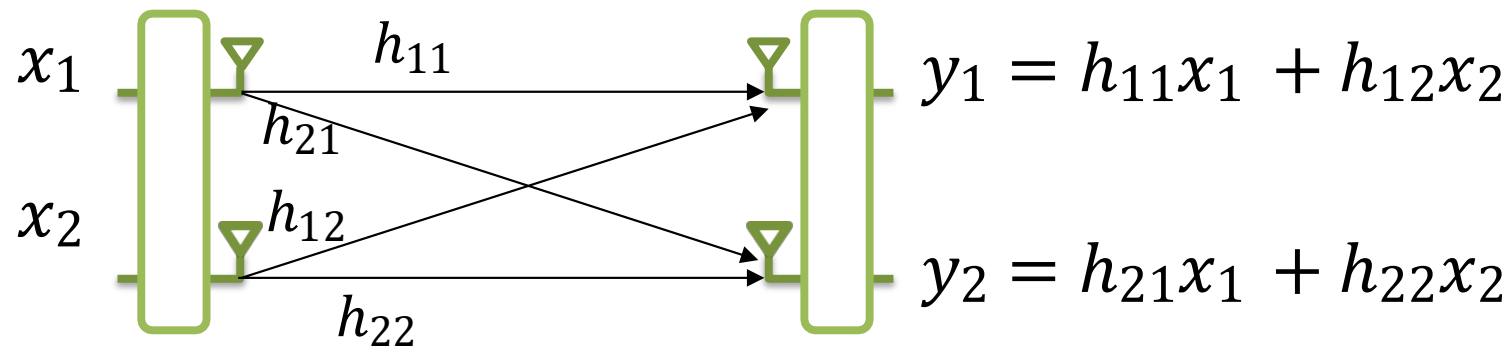
**For N antennas, $\mathbf{H}$ is
 $N \times N$ matrix $\rightarrow O(N^3)$**

Estimate $\mathbf{H}$, compute $\mathbf{H}^{-1}$ and invert the channel!

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Noise amplification

MIMO: Vector Representation



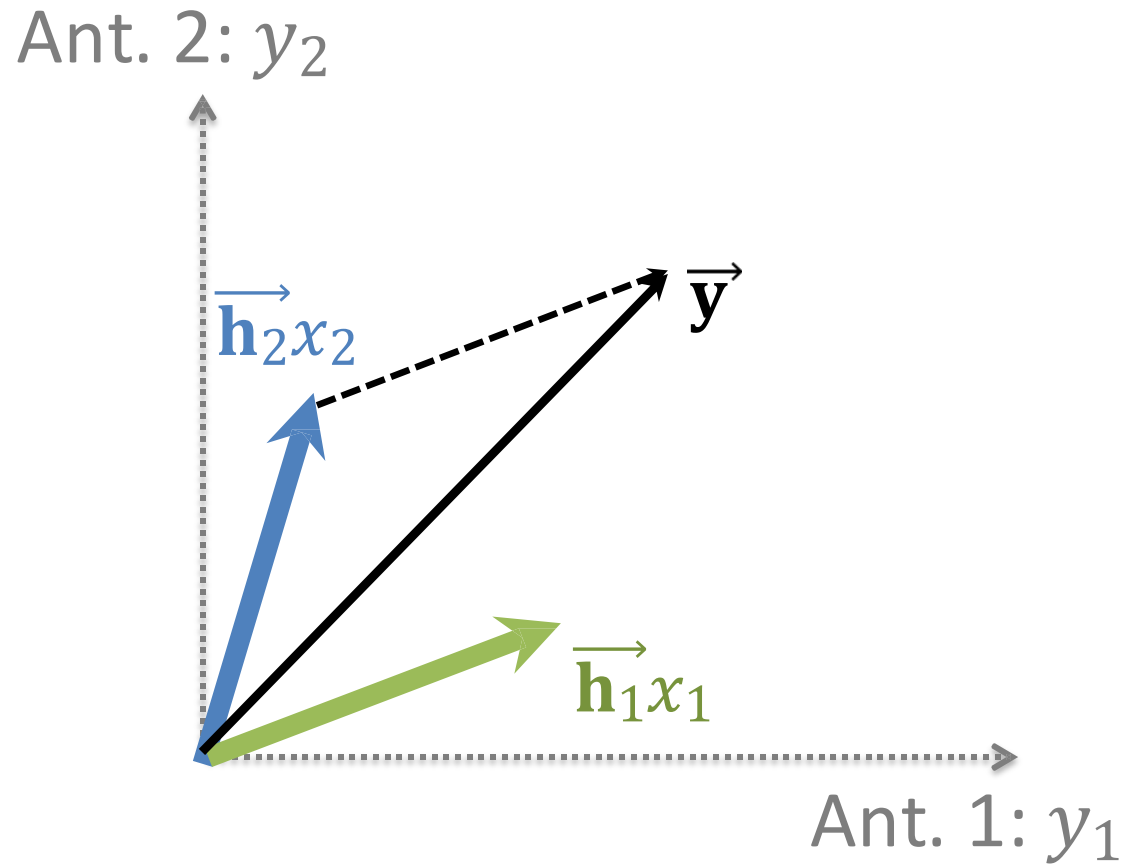
$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

How to recover x_1 and x_2 ?

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2 = \overrightarrow{\mathbf{h}}_1 x_1 + \overrightarrow{\mathbf{h}}_2 x_2$$

MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$



MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_1 , project on a vector $\vec{h}_2^\perp$ orthogonal to $\vec{h}_2$

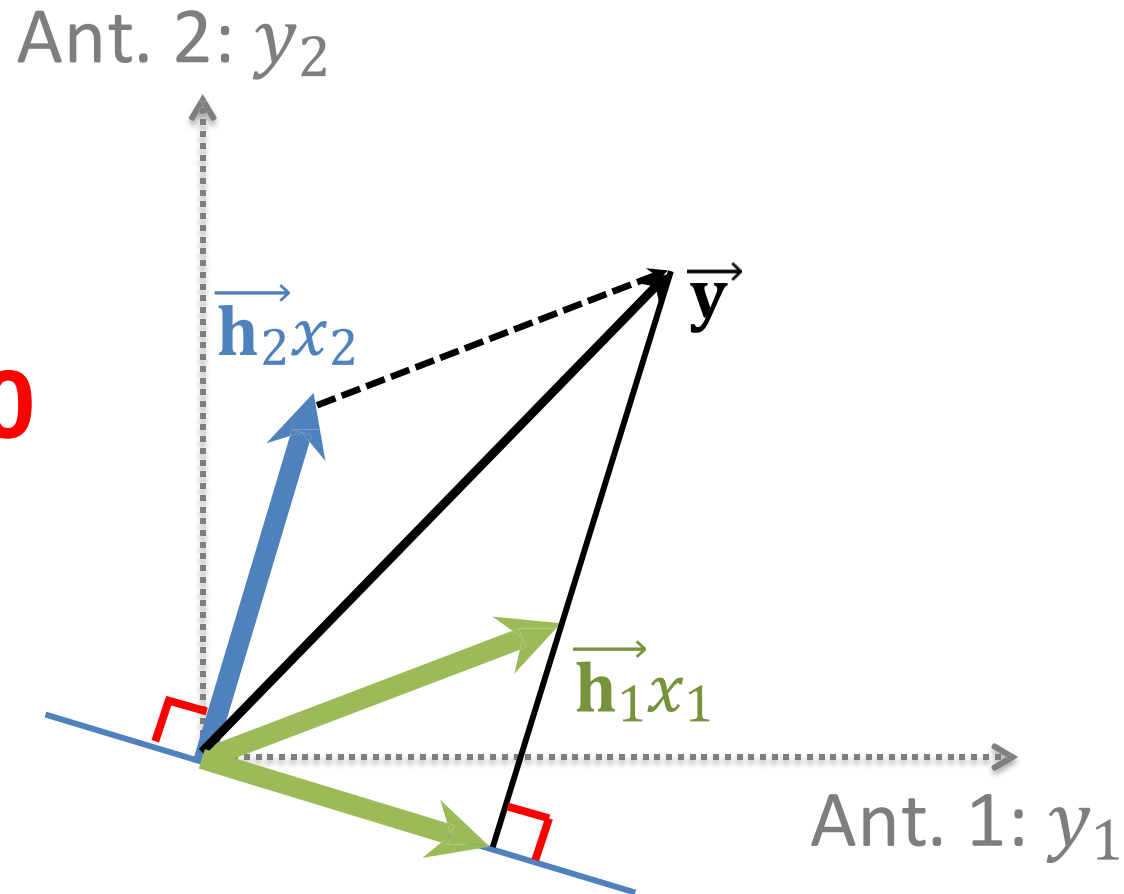
$$\vec{h}_2^\perp \vec{y} = \vec{h}_2^\perp \vec{h}_1 x_1 + \vec{h}_2^\perp \vec{h}_2 x_2$$

$$= \vec{h}_2^\perp \vec{h}_1 x_1$$

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_2^\perp = \begin{bmatrix} h_{22} & -h_{12} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_2^\perp \vec{y} &= h_{22} h_{11} x_1 - h_{12} h_{21} x_1 \\ &\quad + \cancel{h_{22} h_{12} x_2} - \cancel{h_{12} h_{22} x_2} \\ &= (h_{22} h_{11} - h_{12} h_{21}) x_1 \end{aligned}$$



MIMO: Antenna Space

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2$$

To decode x_2 , project on a vector $\vec{h}_1^\perp$ orthogonal to $\vec{h}_1$

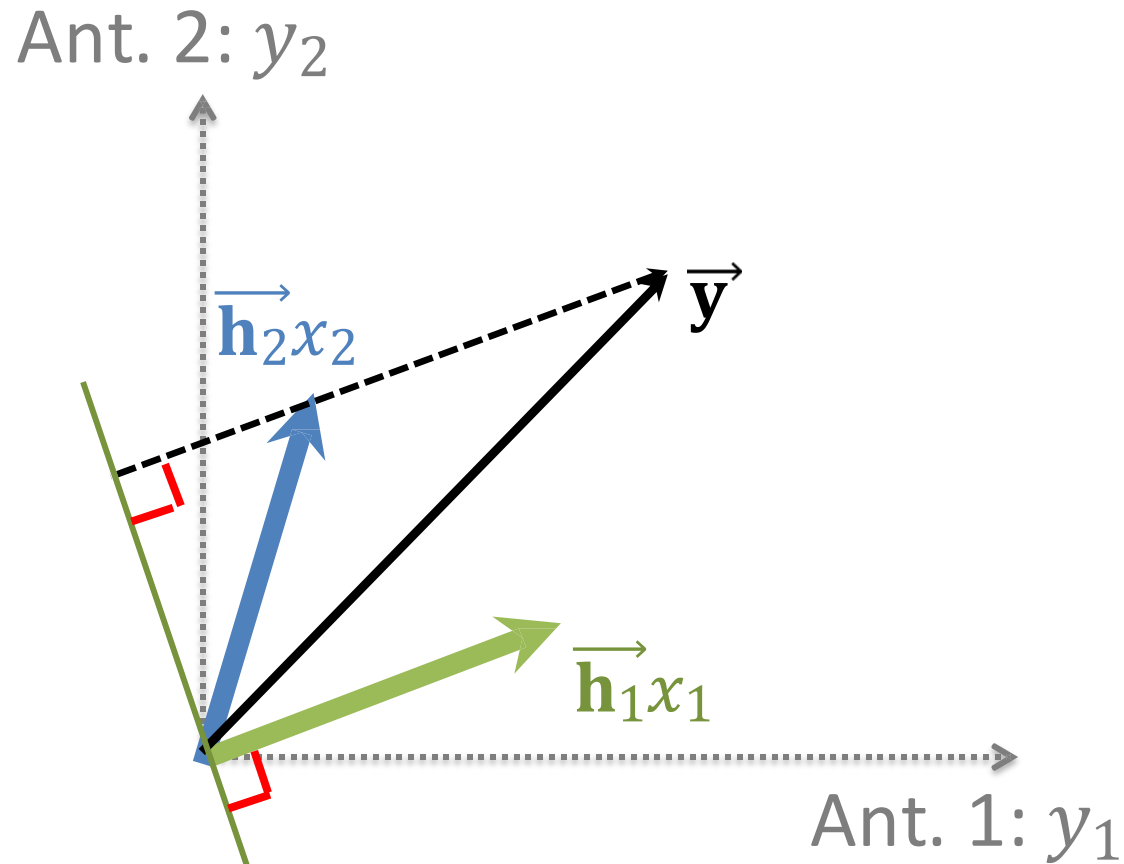
$$\vec{h}_1^\perp \vec{y} = \vec{h}_1^\perp \vec{h}_1 x_1 + \vec{h}_1^\perp \vec{h}_2 x_2$$

$$= \vec{h}_1^\perp \vec{h}_2 x_2$$

$$\mathbf{y} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1 + \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

$$\vec{h}_1^\perp = \begin{bmatrix} h_{21} & -h_{11} \end{bmatrix}$$

$$\begin{aligned} \vec{h}_1^\perp \vec{y} &= \cancel{h_{21} h_{11} x_1} - \cancel{h_{21} h_{11} x_1} + h_{21} h_{12} x_2 - h_{11} h_{22} x_2 \quad \mathbf{0} \\ &= (h_{21} h_{12} - h_{11} h_{22}) x_2 \end{aligned}$$

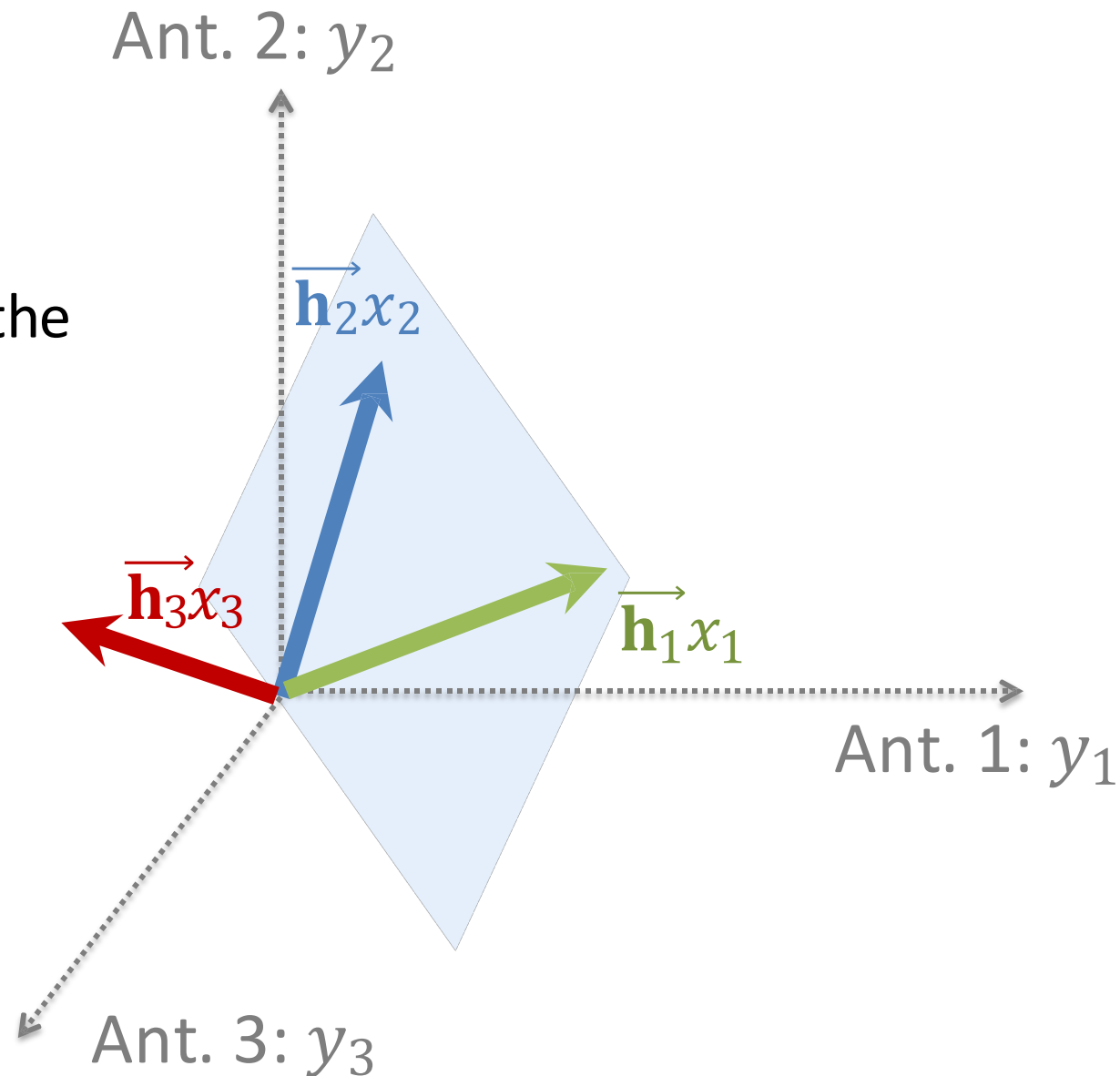


MIMO: Antenna Space

What about MIMO with more antennas?

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2 + \vec{h}_3 x_3$$

To decode x_1 , project on a vector $\vec{h}_{23}^\perp$ orthogonal to the plane formed by $\vec{h}_2$ and $\vec{h}_3$

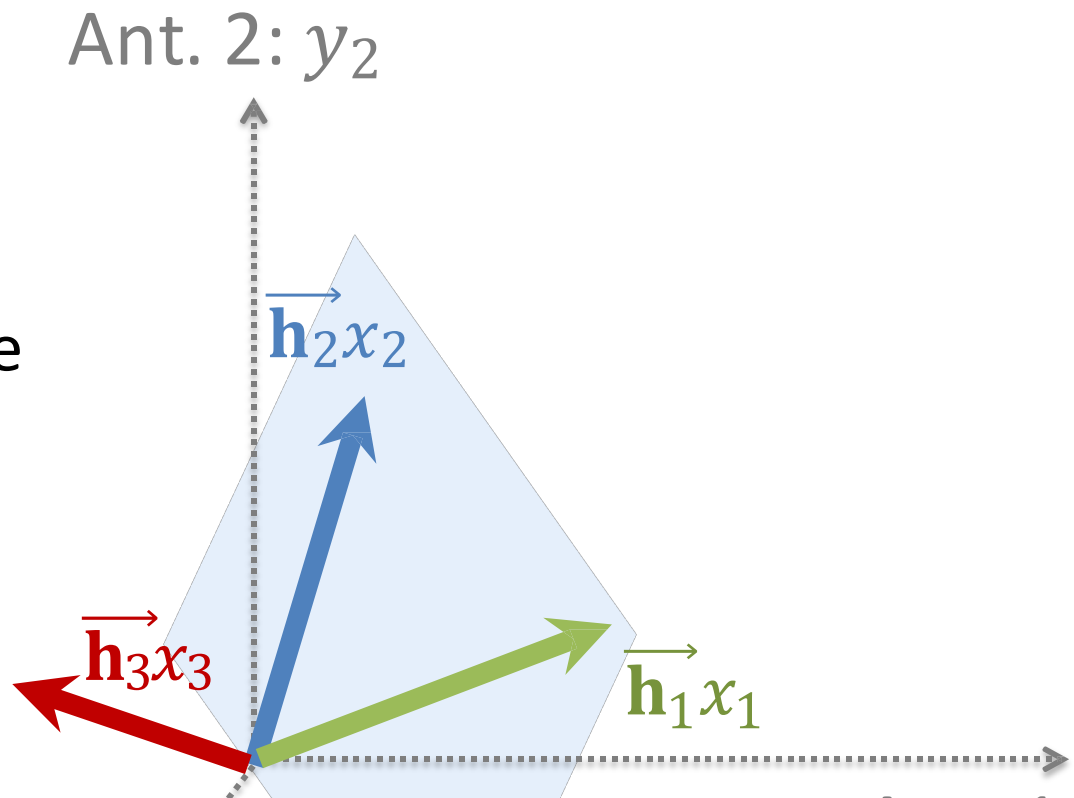


MIMO: Antenna Space

What about MIMO with more antennas?

$$\vec{y} = \vec{h}_1 x_1 + \vec{h}_2 x_2 + \vec{h}_3 x_3$$

To decode x_1 , project on a vector $\vec{h}_{23}^\perp$ orthogonal to the plane formed by $\vec{h}_2$ and $\vec{h}_3$



N antenna MIMO, receives signals in N dimensional space
→ Can decode N parallel signals